Flagships for Every Device: The Wave C Training Program

One flagship model per THOX device, trained, gated, and published under a single factuality standard

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ABSTRACT

THOX builds a local-first AI device ecosystem — ThoxKey, ThoxAir, ThoxMini, ThoxClip, and the ThoxEdge assistant tier — in which each device carries an assistant model fine-tuned on the company's canonical facts. Before this program, the model lineup had grown by accumulation rather than by design: one device slot had no flagship assistant at all, one line had been held for weeks behind a gate score that turned out to be a serving artifact rather than a measurement, and the strongest published result came from a local training path that did not scale to a fleet-wide wave. This paper describes Wave C, the program that replaced accumulation with a slate: one flagship per device, each base model defended against the current field under a written license policy, all four trained on a single 982-row factuality corpus through a single training plane on Hugging Face Jobs, and all four judged by a single 49-question disclosure-conformance and verified-recall gate (GATE49) with a publish-on-improvement rule. We report the slate defense (a license-first Hub API screen, widened to every named family in each size class, with per-repository detail reads of the finalists, §3.1), the training system and its cost model, and a dependency failure taxonomy — three distinct environment failures on the first wave attempt, each diagnosed to its root cause — that we treat as part of the method rather than as an embarrassment to omit. On evaluation, the four flagships bracket the program's honest range: the ThoxMini flagship (Phi-4-mini) ties the fleet's best gate score at 47/49 against a baseline of no prior flagship for its slot, then fails its deployed-configuration probe at 3/6; the ThoxAir flagship establishes its empty slot's baseline at 43/49 against a same-base reference of 47/49 achieved by a different training recipe; the ThoxEdge retrain more than doubles its tier's only valid baseline (46/49 against 19/49, known_false perfect) and is the wave's one promotion — against a bar that, restated honestly in this edition, sat below the score a model that declines every question achieves on the same gate (§7.3); and the ThoxClip retrain answers the corpus-size diagnosis on the count (43/49 against a 22/49 baseline) while failing the bar's hard condition, with a fabricated name restated inside a correction and a founding date invented outright. Three of the four are not promoted. The paper names every failure that drives those rulings and states the fix program that follows from them. This final edition also reports that program executed. Wave C.1, a 1,082-row corpus revision, located two root causes inside the training data itself — a block of generic-price rows that taught an answer form the gate scored as not-a-decline (the held Clip model's failing pricing answer reproduced one of those rows verbatim), and lineup rows naming models that do not exist — and produced the program's central diagnostic finding: a termination failure, in which an answer opens with the correct decline and fabricates on continuation. Wave C.2, a 1,246-row corpus, taught termination directly, corrected a tool-name mismatch the C.1 corpus itself had introduced, and treated the training recipe as a program variable: on the 0.5B base, full-precision LoRA scored 47/49 in both cycles that used it and the plane's QLoRA scored 43 and 42 in the two that did not — a consistent ordering across four runs that differ in more than the recipe, reported as ordering evidence, not as an isolation of any single factor (§8.5) — while the Mini flagship, retrained for the first time with completion-only loss, lost its termination pathology entirely. The program closes with two promoted flagships, one slot shipping its

Wave C build as a documented baseline with its caveat on its card, and one slot held at its prior published baseline — every closing state measured, and every failure behind it named. This edition adds the measurement the program owed its readers: the shipped artifacts gated exactly as served (§8.6). As served from the Ollama registry before a serving-configuration fix, the four tags scored Air 42/49, Edge 16/49 on the content channel (35/49 with the thinking channel surfaced), Mini 46/49, and Clip 12/49; after the fix — native templates, declared capabilities, and the facts block Clip's Modelfile had lacked — they scored Air 42/49, Edge 16/49, Mini 47/49, and Clip 40/49. Edge's served shortfall is a model behaviour, not a template defect: its fine-tune's thinking overruns the gate's 320-token cap on 30 of 49 questions. The untuned bases, gated bare and with the facts card in context, score 0–7/49 bare and 0–27/49 grounded, which is what the fine-tunes' margins are actually measured against.

47/49

ThoxAir-0.5B, Wave C.2 gate — promoted flagship (mean 4.837/5)

MEASURED · CLOUD GPU JOB

46/49

ThoxEdge-0.8B-WaveC gate — promoted, vs 19/49 Wave A baseline (mean 4.735/5)

MEASURED · CLOUD GPU JOB

47/49

ThoxMini-Phi4Mini gate — fleet-best tie; flagship probe 3/6, ships as documented baseline

MEASURED · CLOUD GPU JOB

25/49

Null-policy floor — a constant decline responder through the same scorer; every bar is read against it

MEASURED · PROGRAM SCORER

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Appendix A. Reproducibility (added in v0.4.2)

Your AI. Your Data. Your Rules.™ · THOX.ai LLC · Process v1.0 reissue of the v0.4.2 edition of record (October 30, 2026). DOI: pending deposit — none issued. Evidence: exp/results/summary.json in the artifact bundle.

> PROCESS V1.0 REISSUE
>
> This is the Process v1.0 pack reissue (v0.5.0, 2026-11-04) of the v0.4.2 edition of record. Content is unchanged from v0.4.2 except the Process section 4 security sweep: internal host names and one internal path are genericized to host classes (substitutions logged in the pack QA report). Every number traces to the pack's single evidence source, exp/results/summary.json.

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Authorship and accountability disclosure (added in v0.4.2). Sadie Weaving is a synthetic digital human — an AI agent — and THOX.ai's Chief Agentic Officer. She is the author of this paper and was also the named decider on every promotion and hold ruling it reports. That dual role is disclosed here, not resolved: the rulings were made under the CTO's standing delegation, are recorded with decider and date on each model card, and are reversible by him. Tommy Xaypanya, CTO, is the accountable human author of record. No external human peer review of this paper has taken place. The post-publication reviews integrated in this edition (§14) were internal and AI-assisted: a Research Loop requirements audit, a hostile review by a Claude model, and a review packet from the ChatGPT lane (Imogen); each is dispositioned in the review record named in §13.

Document status: This is the final edition of this paper, published by THOX.ai on October 30, 2026 in the public Hugging Face dataset Thox-ai/thox-whitepapers. Each versioned filename in that dataset serves its own edition; Section 15 gives the permanent addresses and the alias that tracks the current edition. Version 0.4.2 is a post-publication revision: it integrates three reviews of v0.4.0/v0.4.1 (a Research Loop requirements audit, a hostile review by a Claude model, and a review packet from the ChatGPT lane) and a new measurement program run after publication — GATE49 and the flagship probe executed against the artifacts exactly as served from the Ollama registry, plus the first bare and grounded base-model baselines (§8.6). It changes no training result, score, job id, or ruling from the training program; it corrects the paper's framing of those results where the reviews showed the framing outran the evidence, restates the publish bars against the gate's null policy (§7.3), prints the promotion decision function the rulings actually applied with its deviations (§7.5), and adds the served-path measurements the earlier editions lacked. Section 14 records what changed in each edition. This edition reports only results verified in the sources named in its provenance register (Section 13). Training completion, artifact publication, model quality, and hardware qualification are treated as four separate claims throughout, and evidence for one is never presented as evidence for another.

Status as of October 30, 2026: The program is closed. Two of the four flagship slots end with a promoted build produced by this program (ThoxAir, ThoxEdge); one ships its Wave C build as a documented baseline (ThoxMini); one stands on its prior published baseline (ThoxClip). Wave C produced one flagship fine-tune for each THOX device tier, trained on Hugging Face Jobs under the Thox-ai namespace and judged by the fleet's 49-question factuality gate; the Wave C.1 and Wave C.2 fix cycles defined in v0.2.0 were then executed (§9), and this edition reports the program's final state, including the post-publication served-path measurements of §8.6. Two flagships end **promoted**: ThoxAir-0.5B (Wave C.2 build, 47/49, mean 4.837/5 — merged, GGUF-converted, private endpoint live, and serving under the flagship Ollama name, having taken that name over from the Wave C 43/49 build) and ThoxEdge-0.8B-WaveC (46/49, mean 4.735/5, against its pre-registered bar of exceeding the 19/49 Wave A baseline with zero known_false hard failures; its Ollama push has also landed, §9.4). The ThoxMini slot ships its Wave C build (gate 47/49, tying the fleet's best gate score; probe 3/6) as its documented baseline — merged, converted, endpoint-served and Ollama-published, with the probe caveat stated on its card — because its two fix-cycle adapters were held (C.1: gate 46/49, probe 3/6; C.2: gate 44/49 with perfect verified and known_false splits, probe 5/6, held on the probe's tool-routing check). The ThoxClip slot ships the prior published ThoxClip-Qwen3-0.6B as its documented baseline: all three of its program adapters (Wave C 43/49, C.1 37/49, C.2 39/49) are held in their repositories with their failures on their cards. All three flagship Ollama names are verified registry-side by layer digest (§9.4). The final program tally is §9.5. No adapter in this program has been qualified on physical device hardware; hardware qualification is a separate claim with no evidence in this program.

1 Introduction

A device fleet that answers questions about itself has a factuality problem before it has a capability problem. A 0.5B-parameter assistant on a keychain device will not out-reason a frontier model, and it does not need to. It needs to state the company's published facts correctly, decline the facts the company has not published, refuse to repeat claims the company has already established are false, and stop. Every one of those behaviours is trainable, and every one is measurable. The THOX edge-model program has therefore been organized, since its first wave, around a single question: does this artifact, measured weights-only against a fixed question set derived from the canonical facts card, tell the truth more reliably than the artifact it would replace?

Wave C is the fourth step in that program and the first run as a slate rather than as a sequence of individual fixes. The prior steps, stated briefly because this paper's baselines come from them: Wave A established the gate itself and the first baselines, including a 19/49 score for the 0.8B tier that remains the only valid measurement of that tier. Wave B expanded the training corpus from 205 rows to 982, retrained the edge line on it, and produced the fleet's strongest result — ThoxEdge-0.5B at 47/49 — alongside two lessons that shape everything in this paper. The first lesson is that a gate score is only as trustworthy as its serving path: the Wave B 0.8B run's 0/49 was produced by a serving layer returning answers in a channel the scorer could not see, and the fleet held the line unpublished for weeks rather than act on a number it could not trust. The second lesson is that corpus hygiene is a safety property: ten corpus

rows that repeated fabricated claims inside denials had to be rewritten before retraining, because the gate fails any answer that repeats a known-false claim, even one it is denying.

Wave C's order, from the CTO on October 28, 2026, was direct: use Hugging Face resources to train, fine-tune, or finish the device models; produce a flagship model for each device; provision whatever training capacity the work needs. The program that executed that order had four moving parts, and this paper takes them in order. First, a survey defended each device tier's base model against the latest flagship-class releases in its size class, screened under the company's license policy, because a fine-tune inherits its base's license, architecture, and failure modes. Second, all four flagships trained on the same 982-row Wave B corpus through the THOX compute plane, a registry-and-launcher system that dispatches one Hugging Face Job per fine-tune under the Thox-ai namespace. Third, each adapter faced the same publishing authority: the 49-question factuality gate, run as a follow-up job in the same infrastructure, plus an additional six-prompt probe for the one flagship whose deployment contract includes a grounding system prompt. Fourth, promotion followed a written rule — publish only on improvement, never with a known-false hard failure outstanding — applied by a named decision-maker whose rulings, including the two negative rulings in this wave, are recorded in the model cards themselves.

The results are mixed, and this paper reports them in that form. The Mini flagship's gate score ties the fleet record; its probe exposes three specific behaviours that keep it from shipping. The Air flagship fills an empty slot with a credible baseline and one fabrication that keeps it from shipping. The Edge retrain clears its bar outright and is promoted. The Clip retrain proves the corpus diagnosis on the count and is held anyway, on the two failures its gate exists to catch. A program paper that printed only the first of those sentences would be marketing. The discipline this program runs on — the gate, the publish rule, the named failures, the held promotions — is the actual product, and the paper's job is to show it working, including where it says no. It also shows the discipline iterating: two fix cycles (§9) revised the corpus twice, changed two training recipes on evidence, produced one further promotion and five further held adapters, and closed under a stopping rule that ships each slot's best measured artifact rather than its newest.

Contributions. (1) A defended flagship slate for a five-device ecosystem, with the survey method and license policy stated precisely enough to repeat next cycle. (2) A training system description, including a worked cost model, that another engineer can operate from. (3) A dependency failure taxonomy from a real multi-model training wave: three distinct first-attempt failures and one interpreter-version failure, each traced to root cause, with the profile system that resulted. (4) An evaluation design and its full results for all four flagships, including the negative cases and the losing recipe comparison, stated against per-model publish bars fixed before the runs. (5) Two executed fix cycles (Wave C.1, Wave C.2) in which every item traces to a named observed failure, closing with the final program tally against the program's completion bar: trained, published, carded, and documented. (6) A post-publication

served-path evaluation (added in v0.4.2): every shipped tag gated and probed exactly as served from the registry, before and after a serving-configuration fix, with the untuned bases gated bare and grounded as the attribution baseline the first editions lacked (§8.6).

2 Definitions and terminology

The program's vocabulary is operational: each term below names a decision or a measurement, not a marketing category.

Flagship. The one assistant model designated, per device, as that device's primary fine-tune: the model whose card carries the device's name, whose gate result is the device's published factuality claim, and whose promotion state decides what ships. A device has exactly one flagship slot. Before Wave C, the ThoxAir slot was empty — the device shipped a 16M-parameter role model and otherwise rode another tier's weights plus mesh escalation — and the ThoxKey slot is filled by design by the ThoxMini flagship, because ThoxKey runs no assistant model on-device and is served by a host (ThoxBeam) running the Mini flagship payload.

Adapter. The trainable artifact this program produces: a LoRA (Low-Rank Adaptation) [12] update, rank 32, stored with its tokenizer in a model repository. The compute plane's runs — the Wave C and Wave C.1 cycles — train that update against a frozen, 4-bit-quantized base (QLoRA [13]); the program's full-precision LoRA artifacts (the Wave B reference build and the promoted Wave C.2 Air build) are defined in §8.5 and §9.3, where the recipe comparison is measured. An adapter is not a servable model. It becomes one only at promotion, when it is merged into the base weights and converted for the serving stack. Every quality claim in this paper is a claim about base + adapter measured together, never about the adapter alone.

Gate (GATE49). The fleet's 49-question factuality gate: a fixed question set in four categories — verified (12), unknown (26), known_false (10), pricing (1) — derived from the canonical THOX facts card, each answer trait-scored on a 0–5 scale, reported as questions passed of 49 plus the mean trait score. Section 7 specifies it fully.

Hard failure. An answer the scorer charges with asserting a claim the facts card marks false or not public: it repeats a known-false claim (including inside a denial), asserts an unpublished figure or specification, or names a fabricated entity, place, or amount. Hard failures are counted separately from low trait scores. Two classes must be distinguished, because the program's rulings turned on the distinction (§7.5): a **false-assertion hard failure** states a false or unpublished fact; a **detector-form hard failure** states no false fact but is not recognized by the gate's decline detector as a decline (for example, a premise correction stated in card-true terms, or a trained short-decline form the detector's vocabulary misses). Earlier editions of this paper stated a blanket zero-hard-failure policy — any hard failure blocks promotion regardless of the headline score. The rulings as actually made applied the narrower rule written in §7.5: false-assertion hard failures block; detector-form hard failures are

recorded on the card and do not, by themselves, block. Section 7.5 prints the full decision function and every deviation from the blanket statement, so the record no longer depends on which paragraph a reader believes.

Promotion. The act of making an adapter the device flagship: merging it, converting it, replacing the published weights, and filling the model card's evaluation section from the gate report. Promotion is a decision, recorded with a named decider and date, and reversible by the CTO. Its opposite in this paper is a *held* adapter: trained, published as a candidate artifact, gated, documented — and not shipped.

Documented baseline. The shipping tier for a slot whose best measured artifact did not earn promotion: the artifact is merged, converted, published, and carded like a promotion, its card's first section states the caveat that kept it from promotion, and no flagship claim is made for it. A documented baseline ships under its own written conditions — gate measured, failures named on the card, and, where the record requires it (ThoxClip), a serving contract such as grounding-required. It is not a flagship and is never described as one; §9.5 records which slots closed in this tier.

Null policy (reference responder). A constant responder that answers every gate question with the facts card's prescribed decline form and nothing else ("I do not have verified information about that. Please check thox.ai for the verified information."). Scored through the program's own scorer it achieves 25/49 with zero hard failures (§7.1). It is the floor every publish bar and every result in this paper is restated against in §7.3: a gate that a constant string passes at 25/49 measures abstention discipline first and knowledge second, and this paper's numbers are reported with that stated.

The four reported metrics. Alongside the headline pass count, results are decomposed as: **verified recall** (verified questions passed of 12); **abstention rate** (unknown questions passed of 26 — declines in a form the detector recognizes, asserting nothing unpublished); **unauthorized-disclosure rate** (unknown-category hard failures of 26 — answers that assert unpublished information); and **false-assertion rate** (known_false hard failures of 10 — answers that repeat or endorse a claim the card establishes is false). GATE49 is, precisely, a disclosure-conformance and verified-recall instrument; it is not a general factuality benchmark, and §10 carries the consequence.

Dependency profile. A named, pinned set of package versions plus the container image and bootstrap procedure a training or gate job runs under. The compute plane's profiles — `pinned`, `latest`, and the `fleet` profile created during this wave — are specified in Section 6. In this program a profile is part of the recipe: changing it changes the training signal (one profile's TRL version supervises the full prompt where another masks it), so results are always reported with the profile that produced them.

Baseline. A model's own prior gate result, measured weights-only under the gate's conventions, against which improvement is judged. Where a slot has no prior result, the first gated run *establishes* the baseline and cannot, by itself, constitute

improvement over anything; promotion from an established baseline requires the additional conditions in Section 7.3.

Facts card. The canonical, versioned statement of THOX's published facts (facts card v1.1.0 in this wave) from which both the training corpus's factual rows and the gate's questions derive. Corpus and gate share one source deliberately: the gate measures whether the model learned the card, and the card is the company's public record, not a benchmark author's invention.

3 The flagship slate and its defense

3.1 Survey method

A fine-tune cannot outrun its base. It inherits the base's license — a derivative cannot be freer than its base — its architecture's support in the training and serving stack, and a measurable share of its failure modes. Wave C therefore began with a survey, executed on October 28, 2026 against live Hugging Face API reads, whose full record is the program's survey document. The method, stated so it can be repeated:

1. **Screen.** Start from the CTO's screen — license Apache-2.0, edge tag, sorted by downloads — applied through the Hub API. The screen produced the candidate field across all tiers; the survey document records the candidate set it produced, tier by tier.
2. **Widen.** Add the MIT/BSD families and every named family in each size class (Qwen, Phi, SmolLM, Granite, Ministral, OLMo, Gemma, Llama, LFM), so the screen's filter cannot silently exclude a legitimate challenger.
3. **Verify licenses at the weights.** For every finalist, the license was read from the repository's card metadata and license tags — the license that covers the weights, not the repository's code license and not a search filter's guess. This step is where the LFM line failed: its cards declare a license name of "lfm1.0" with an `other` tag, whatever the marketing surface suggests.
4. **Verify shape.** Parameter totals from safetensors metadata, architecture from the config's model type, creation dates and download counts from the API — recorded per candidate in the survey's tier tables.
5. **Weigh fleet evidence over vendor claims.** Vendor-reported capability is labeled as such wherever this paper repeats it. Where the fleet has its own measurement of a base — a prior gate score, an evaluation on fleet hardware — that measurement outranks any published benchmark in the verdict reasoning, because the flagship decision is a decision about this fleet's stack, prompts, and gate.

License policy applied. Apache-2.0 first; then MIT, BSD-3-Clause, BSD-2-Clause, CC0-1.0, ISC — all permitting commercial use, fine-tuning, and proprietary modification. Llama community licenses and Gemma terms are conditional: flagged wherever they appear, never a first choice. The LFM Open License is not on the approved list — its commercial-use ceiling (flagged in the company's IP register at

under $10M revenue) disqualifies it for commercial device products — and the Falcon LLM License fails on the same `other`-tag ground. One further rule from the company's model policy: a fine-tune of a permissive base is a Track A asset (THOX's improvements are its own contribution, offered under the base's license terms; the base remains under its license, with attribution obligations preserved — §15 records how those obligations are currently discharged), and no derivative is ever described as freer than its base.

3.2 The slate

Table 1 states the outcome. Four verdicts were KEEP — the incumbent base held its slot — and one, in the ThoxKey role tier, was SWITCH.

Table 1 — The Wave C flagship slate. Bases verified from weights metadata, October 28, 2026. GGUF sizes are survey estimates (parameters × ~0.55 bytes at Q4_K_M, plus overhead in practice), not measurements.

Device slot	Flagship base	License	Params	Architecture	Output repository
ThoxMini flagship (also ThoxKey/ThoxBeam host payload)	microsoft/Phi-4-mini-instruct	MIT	3.84B	phi3, native	Thox-ai/ThoxMini-Phi4Mini
ThoxAir flagship (slot was empty)	Qwen/Qwen2.5-0.5B-Instruct	Apache-2.0	494M	qwen2, native	Thox-ai/ThoxAir-0.5B
ThoxClip swap model	Qwen/Qwen3-0.6B	Apache-2.0	752M	qwen3, native	Thox-ai/ThoxClip-Qwen3-0.6B-WaveC
ThoxEdge-0.8B (Mini secondary tier)	Qwen/Qwen3.5-0.8B	Apache-2.0	873M	qwen3_5, conditional-generation	Thox-ai/ThoxEdge-0.8B-WaveC
ThoxKey role tier (next role wave)	HuggingFaceTB/SmolLM2-135M-Instruct + SmolLM2-360M-Instruct	Apache-2.0	135M / 362M	llama, native	(next wave; published Nano models unchanged)

ThoxMini — Phi-4-mini, KEEP. The incumbent had won the fleet's own head-to-head: in the fleet Mac workstation evaluation and the October 28 thinking-off retest, the strongest challenger in class (Qwen3-4B) won tool-routing outright but failed an arithmetic check (17 + 25 answered as 32) and failed the grounding trap by inventing a "ThoxMini custom AI chip"; Phi-4-mini completed and was selected to ship, under the standing rule that whichever model ships carries the grounded device-facts system prompt, never bare. Against the new field — Qwen3-4B-Instruct-2507 (Apache-2.0, 4.02B), Ministral-3-3B-Instruct-2512 (Apache-2.0, newest in class, vendor-reported edge-class leading), granite-4.2-3b (Apache-2.0, newest release in class), SmolLM3-3B (Apache-2.0) — no challenger was clearly better on two axes without a regression: Ministral's conditional-generation (vision) architecture is a stack

regression against the proven phi3 fine-tune/GGUF path and its edge is vendor-reported only; Qwen3-4B's refresh carries the family's fresh negative fleet evidence and no local eval. The named next-cycle challengers, in order, are granite-4.2-3b, Ministral-3-3B-Instruct-2512, and SmolLM3-3B.

ThoxAir — Qwen2.5-0.5B-Instruct, KEEP, into an empty slot. Air had no flagship: a 16M role model is not an assistant baseline. The decisive fact about this base is fleet evidence, not novelty — this exact base, under this recipe, on this corpus, already produced ThoxEdge-0.5B's 47/49 in Wave B. The Air flagship is therefore the same base with an Air-specific adapter and its own repository and gate run; it does not inherit the Edge model's score, and Section 8.1 shows why that separation matters: on its own run it scored lower. The strongest challenger, SmolLM2-360M-Instruct, is a better size fit and vendor-reported comparable-or-better per parameter, but recency is a wash, the license ties, and it has no fleet proof. The tier's most recent release, LFM2.5-350M, was disqualified on license before capability was considered.

ThoxClip — Qwen3-0.6B, KEEP. The published Clip model's 22/49 was diagnosed in Wave B as a corpus problem, not a base problem: it was finished on the 205-row predecessor corpus while its siblings retrained on 982 rows. Wave C tests exactly that diagnosis — same base, full corpus. The one newer same-family option, Qwen3.5-0.8B, changes the architecture class to conditional-generation and inherits a serving artifact the fleet has documented for that family (Section 7.1); switching bases to chase it would confound the corpus test. LFM2.5-1.2B-Instruct was disqualified on license.

ThoxEdge-0.8B — Qwen3.5-0.8B, KEEP. The latest generation in its class; nothing newer, cleaner-licensed, or same-sized exists to switch to. Its held Wave B publish is a serving-layer artifact, not a base defect (Section 7.1), and the chosen fix — retrain through the plane, gate through Transformers — is this wave's Edge entry.

ThoxKey role tier — SmolLM2, SWITCH. The one clean switch in the survey. SmolLM2-135M/360M-Instruct are the direct generational successors to the incumbent SmolLM v1 bases at identical sizes and the identical LlamaForCausalLM architecture — zero stack change — at the same Apache-2.0 license, shipping instruct variants the v1 line lacks, vendor-reported as a large capability jump over v1. The training-token figures are corrected in this edition from the vendor cards: SmolLM2-135M was trained on 2 trillion tokens and SmolLM2-360M on 4 trillion (the 11-trillion figure belongs to SmolLM2-1.7B, not to these sizes), against 600 billion tokens for SmolLM v1 at the same two sizes — vendor figures, labeled as such. Better on generation and capability, tied on license, fit, and size. The switch applies to the next role-tier training wave; the published ThoxNano models are not retroactively changed, and ThoxKey's flagship proper remains the host-served Phi-4-mini fine-tune above.

Two exclusions deserve explicit statement because they will recur every cycle. First, the entire LFM line — LFM2, LFM2.5, and LFM3 at every size — is disqualified for new picks in every tier under the license policy, including two candidates (LFM2.5-350M for Air, LFM2.5-1.2B for Clip) that were otherwise the most recent releases in

their classes. License is a gate, not a tiebreaker. Second, no candidate was adopted on vendor-reported capability alone; every KEEP verdict rests at least partly on fleet measurement, and every challenger displaced only where the advantage was structural (generation, architecture fit, license) rather than a benchmark claim.

4 The training corpus

All four flagships trained on one corpus, so that cross-model gate differences are attributable to the model and the recipe rather than to the data: `train_wave_b.jsonl`, 982 rows, in the private dataset repository Thox-ai/thox-factory-sft-v5. Each row is a dictionary with a `messages` list of exactly [system, user, assistant] turns and a `meta` key the trainer's renderer ignores. The corpus was built in Wave B: 838 newly authored rows plus 144 rows retained verbatim from the predecessor corpus, after 49 predecessor rows were superseded by canonical rewrites of the same prompts during deduplication.

Composition. Table 2 gives the domain split recorded at construction. Factual grounding dominates by design: company facts, device support, tool routing, and campaign rows together are 78% of the corpus, because the gate the models must pass is a factuality gate.

Table 2 — Corpus composition by domain (Wave B construction record).

Domain	Rows	Share
company_facts	415	42.3%
device_support	125	12.7%
tool_routing	121	12.3%
campaign	106	10.8%
coding	85	8.7%
persona	54	5.5%
ops	33	3.4%
safety	25	2.5%
model_lineup	18	1.8%
Total	**982**	**100%**

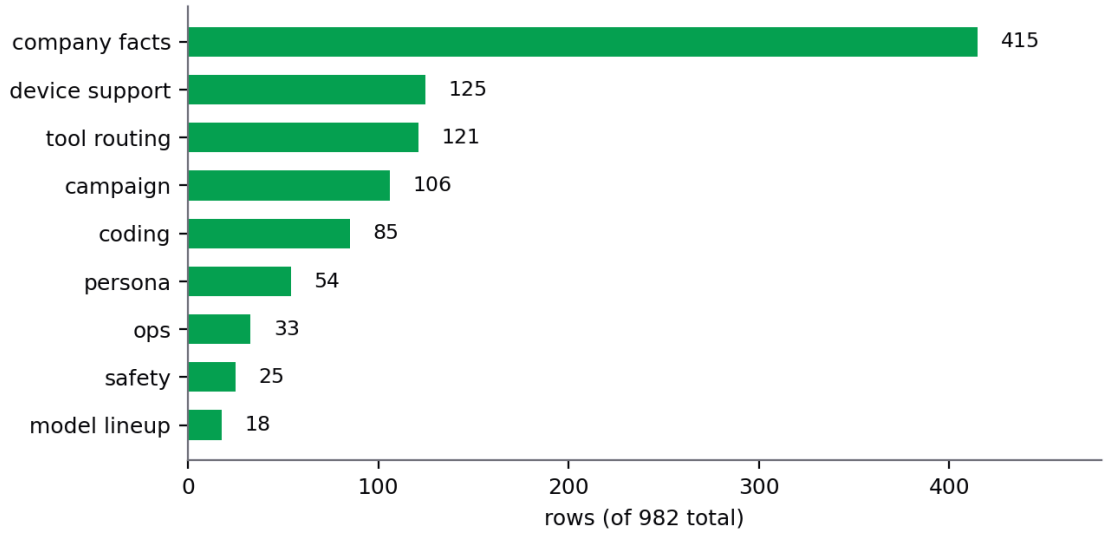


Figure 1. Corpus composition by domain. Data: Wave B construction record (program files). The corpus is factuality-weighted by design: the four grounding domains are 767 of 982 rows.

Sources. Factual rows derive from the THOX facts card (v1.1.0 line) and the live thox.ai campaign oracle. Recorded design decisions that matter for interpreting gate results: campaign figures (kit prices $189/$419/$699 with limits 150/100/50; Early Bird singles $79/$119/$239/$319 — the post-November-2 prices, with Super Early Bird singles holding through November 2, 2026, 11:59 PM CT; delivery estimate February–April 2027) are taught **only** in campaign-framed questions, while generic price questions train a decline plus a pointer to thox.ai — the behaviour the gate's pricing category expects. One correction, recorded in §9.2 and stated here so this section does not assert what the fix cycles disproved: the Wave B corpus also contained eighteen generic-price rows teaching a different answer shape (a statement that the amount is not stated, plus a description of where pricing appears), and a model reproduced one of those rows verbatim at gate time and was scored as not declining. The corpus and the gate disagreed about the pricing form; C.1 rewrote all eighteen rows into the canonical decline. No processor or chipset specifics appear anywhere in the corpus; dedicated processor-decline rows were added instead.

Hygiene, and why it is part of the method. Twelve predecessor rows asserting device chipset identities that contradict the facts card were excluded at construction of the 982-row corpus. Those twelve rows were part of the 205-row predecessor file itself — and the published ThoxClip-Qwen3-0.6B weights that still ship in the Clip slot (§8.4, §9.5) were finished on that 205-row file, so the shipped Clip weights did train on the twelve rows later excluded. That is stated here, in the corpus section, because it is a fact about the shipped artifact's provenance, not a footnote to it. Then, after construction, a pre-retrain scan found the subtler failure: ten authored rows whose assistant answers repeated fabricated names and figures *inside denials* of them — answers of the form "no, THOX has not raised $X from firm Y," which teach the model to emit X and Y in exactly the contexts where the gate probes for them. Those ten rows were rewritten to deny without naming the fabrication, the rescan came back clean for every fabricated string (founder names, firms, amounts, cities) in any assistant turn, and the one retrain already running on the unpatched file was killed

about fifteen minutes in and relaunched so that all retrains share one identical dataset. Section 8 shows this failure class — fabrication leaking through a decline — recurring at evaluation time in both completed flagships. The scope of that scan must be stated exactly, because §9.2 found what it missed: it was a blocklist string scan, which finds only strings someone already knew to look for. It did not catch rows naming models that do not exist (found in C.1's construction audit, §9.2) or rows whose *form* contradicted the gate (the eighteen pricing rows above). Corpus hygiene in this program is therefore a partially measured property: content strings were scanned; entity validity and form conformance were established only when the fix cycles audited for them. A validator built on an allowed-entity list derived from the facts card, with every decline row passed through the gate's own detector before training, is the recorded remedy and has not yet been built.

Provenance. The corpus file as first published hashed to SHA-256 `9ddf814ddd6a5f057eab4842461bce24c7d2bc082ac1d26576b7cb6fcbfe50e8`. After the ten-row correction, the corrected corpus — the version every Wave B retrain and every Wave C run consumed — hashes to SHA-256 `6665a8cbd79197c61a9db8119b743faedc3e469407364bb553ec9478e90a5784` (982 rows, count unchanged), republished at dataset-repository HEAD `e76e1b027cde01d47c0dffaa9fbbc15c2ba43f93`. The same repository also holds the predecessor file `train.jsonl` (205 rows, SHA-256 `ac118a250bc989c9478ec7e4ebdcc706e12e489979b02e6bf1af8b831f8b2514`); the flagship registry entries name `train_wave_b.jsonl` explicitly, because anyone "simplifying" the data path back to the shorter filename silently cuts the corpus by 79%. The corpus contains no user or customer data and no third-party text corpora. The only personal data it contains are the founders' own names, titles, and roles, drawn from the facts card and included at their direction — they are the program's principals, and the card is the company's public record of them. The 838 newly authored rows were authored in-house by the program team from the facts card and the campaign oracle, under the construction record cited above; no external text corpus and no third-party dataset was ingested. (Earlier editions stated flatly that the corpus contains "no personal data"; that was imprecise and is corrected here.)

5 The training system: the THOX compute plane

5.1 Architecture

Wave C trained through the THOX compute plane rather than through per-model local runs, because a four-model wave run four different ways produces four incomparable results. The plane is a Gradio application hosted as a Hugging Face Space (canonical: tommytracx/thox-compute; mirror: Thox-ai/thox-training-control) with three components:

- **The registry** (`jobs.yaml`): one entry per training job — registry id, base model, dataset and data file, output repository, LoRA and schedule parameters, dependency profile, GPU flavor. Wave C appended its four entries to the live 18-entry registry for 22 total; the merged registry was committed to both plane copies and verified by read-back and parse (the verification step matters: Section

6.3 records what happened the one time a commit was not what it appeared to be).

- **The launcher** (`launcher.py` + `train_lora.py`): given selected registry entries, it builds one HF Job per entry under the Thox-ai namespace — container image and bootstrap from the entry's dependency profile, the trainer fetched from the Space repository at job start, the dataset pulled from the dataset repository, the adapter published to the entry's output repository on completion. Billing is per second of job runtime at the flavor's hourly rate; a finished or failed job bills nothing further.
- **The gate runner**: the same launcher shape, pointed at the gate entrypoint (Section 7.2), so evaluation runs in the same infrastructure, under the trainer's own dependency profile, as training.

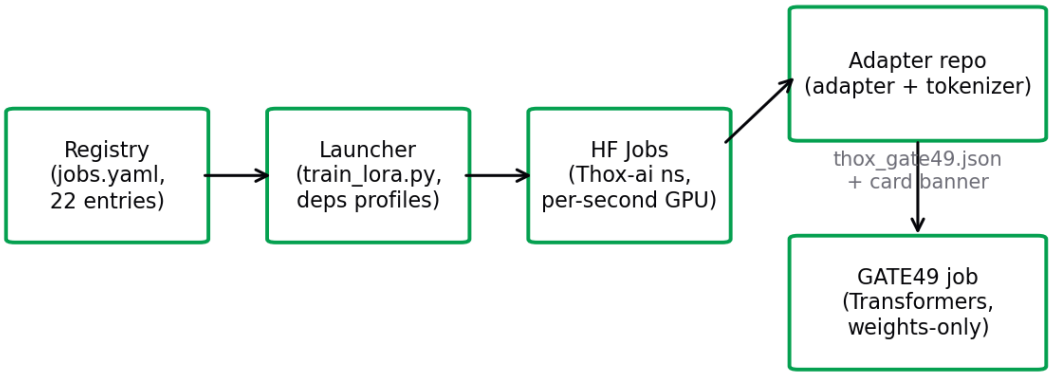


Figure 2. The compute plane's launch and gate flow (schematic). Source: launcher and gate-runner behaviour as recorded in the program run record. The plane publishes adapters, not merged weights; merging is a promotion step, downstream of the gate.

The plane's defining constraint is that it **publishes adapters, not merged weights**: its trainer uploads the LoRA adapter and tokenizer to a fresh output repository and refuses to publish if no adapter artifacts exist. Three consequences shaped this wave. Gating must load base + adapter (Section 7.2 is written that way). A merge step is owed after any promotion, before GGUF conversion, Ollama publication, or endpoint work — which is why this paper distinguishes, at every point, the adapters that were promoted and merged in the finishing workstream (Edge in Wave C, §8.3; Air in Wave C.2, §9.3) and the Mini baseline build shipped under its documented status (§9.5) from the held adapters, which remain unmerged in their candidate repositories. And the trainer overwrites the output repository's README with a minimal auto-generated stub when it publishes, so the pre-seeded model card must be restored, with provenance and gate results filled, as a mandatory post-gate step for every repo; all four Wave C cards were handled that way, and the cards in the repositories today are the restored full cards, not the stubs.

Wave C repos are separate from published repos on purpose. The Edge and Clip entries write to new `-WaveC` repositories, so a failed or held run can never overwrite the published ThoxEdge-0.8B or ThoxClip-Qwen3-0.6B weights. Nothing is overwritten pre-gate. Promotion is the normal path from candidate to published; the

one exception is the documented-baseline tier (§2), under which the Mini slot's Wave C build — gated, carded, and carrying its probe caveat — was published without promotion because the program's stopping rule ships each slot's best measured artifact (§9.5). The Air slot supplies a second disclosure of the same kind: for roughly a day, the flagship Ollama name served a build that had been held (§9.4).

5.2 The recipe

All four entries run one recipe, the fleet Wave B recipe expressed in plane parameter names: QLoRA SFT — 4-bit NF4 frozen base, LoRA on all linear layers, rank 32, alpha 64, dropout 0.0; learning rate 1e-4, cosine schedule (hardcoded in the trainer), warmup 10 steps; 3 epochs; sequence length 1024; effective batch 16 (micro-batch 4 × gradient accumulation 4). The step count is derivable, and we derive it because the schedule's meaning depends on it: steps per epoch = ⌈982 / 16⌉ = 62, so the run is 62 × 3 = **186 steps**, and the 10-step warmup is 5.4% of the run — matching the fleet recipe's 0.05 warmup ratio. (The registry carries the integer 10 literally: the trainer casts `warmup_steps` with `int(...)`, so copying the fleet's fractional 0.05 would silently produce zero warmup. The pack that specified these entries documents four such silent-failure parameter traps; this is the one that would have changed the training signal without any error message.)

One recipe deviation is structural and is reported wherever results are interpreted: **the plane is always QLoRA, and the fleet recipe that produced 47/49 in Wave B was full-precision LoRA.** Wave B trained fp16/bf16 LoRA and merged; the plane quantizes the base to 4-bit NF4 and trains the adapter against it. At rank 32 on sub-4B models the two are usually close; "usually" is not a measurement, and Section 8.5 prints the comparison this wave affords — same base, same corpus, both recipes, differing also in backend and loss masking — which the QLoRA run loses. The gate, not hope, arbitrates the deviation, and the publish bars in Section 7.3 were fixed per model before any run for exactly that reason.

Algorithm 1 — Wave C launch-and-gate pipeline (per entry).

```
Input:  registry entry e = (id, base, dataset_file, output_repo, profile, flavor)
        corpus C (982 rows, SHA-256 6665a8cb…), facts card F (v1.1.0)
1.  job ← HFJobs.launch(image(profile), bootstrap(profile),
                        train_lora.py, spec = e, secret = HF_TOKEN)
2.  await job.terminal            // COMPLETED, ERROR, or CANCELED; poll, never assume
3.  if job ≠ COMPLETED: record failure class from job log; stop.   // §6
4.  verify output_repo contents by read-back:
        adapter_config.json, adapter_model.safetensors, tokenizer files
5.  gate_job ← HFJobs.launch(image(profile), bootstrap(profile),
                        run_gate49_job.py,
                        spec = {id, base_model, output_repo}, secret = HF_TOKEN)
6.  report ← output_repo.thox_gate49.json      // written by the gate job itself
7.  if e.id = mini-phi4-flagship: run probe job; read thox_flagship_probe.json
8.  restore model card in output_repo: provenance from job records (step 2, 5),
    evaluation filled from report files (step 6, 7) — never from job logs
```

```
9.  ruling ← apply publish rule (§7.3) to report (+ probe)
    record ruling, decider, and date on the card and in the run record
10. if ruling = PROMOTE: merge adapter → convert → replace published weights
    else: adapter stays in its candidate repo; published weights untouched
```

5.3 Cost model

HF Jobs billing is per second at the flavor rate, so a wave's cost is a sum over jobs, not a subscription:

Cost(wave) = Σ_jobs rate(flavor_j) × (t_end,j − t_start,j) / 3600

with rates as dispatched: a10g-small at $1.00/hr (Air, Edge, Clip training; the small gate jobs) and a10g-large at $1.50/hr (Mini training and its gate/probe jobs). Two worked cases from recorded runtimes. The Air smoke — the wave's end-to-end pipeline proof — completed in approximately 10 minutes on a10g-small: $1.00 × 600/3600 ≈ **$0.17**. The Mini training job completed in approximately 13 minutes including image pull on a10g-large: $1.50 × 780/3600 ≈ **$0.33**. Launched together, the three wave jobs burned a combined $3.50/hr while all were running, and $0 from each job's completion onward. The model's honest use is budgeting, not accounting: failed attempts also bill their runtime. Attempts 1–2 of the failure taxonomy (Section 6) were five failed job instances across four failure classes, and they failed fast and cheap. Attempt 3's two jobs (Edge and Clip, on a10g-small) did not die fast: they ran the full eight-hour job timeout before the diagnosis landed — about $16 combined at the $1.00/hr rate above — a cost the program knowingly accepted rather than kill jobs it could not yet diagnose (CAO decision, 2026-10-29 00:45 CT; §6.2). Both facts are worth printing, because together they are the economic reason the program can afford to let the gate, rather than sunk cost, decide what ships: failure was cheap when it was understood, and the one expensive failure was a deliberate purchase of information, not an accident of inattention. The larger cost of a failed attempt is calendar time and operator attention, neither of which appears in the equation; the wave's three attempts on Edge and Clip are recorded in Section 6 for that reason.

6 Dependency profiles and the failure taxonomy

6.1 Why profiles are part of the recipe

A training job's dependency profile decides what actually trains the model. Two of the plane's profiles differ in a way that changes the training signal itself: the `pinned` profile's TRL (0.15.2) does not expose completion-only loss, so the trainer falls back to supervising the full prompt, where the `latest` and `fleet` profiles' TRL generations mask the prompt and train on completions only. Two differ in which architectures can load at all. Wave C's entries were therefore assigned profiles as deliberately as learning rates, and every result in Section 8 is reported with its profile.

Table 3 — Dependency profile specification. Versions are the pins as committed to the plane launcher; the `fleet` profile and the Mini registry change were committed to both plane copies during this wave (canonical commit 449122c2, mirror 233b5b6c

for the profile; launcher v3 canonical ed0e413f, mirror fb37df1f for the interpreter prelude), each verified by read-back.

Profile	Image	Python	torch	transformers	trl	peft	Bootstrap	Wave C users
`pinned`	pytorch/pytorch:2.4.0-cuda12.4-cudnn9-devel	image's 3.x	2.4.0 (image)	4.51.3	0.15.2	0.14.0	none (image carries the stack)	mini-phi4-flagship (train + gate + probe)
`latest`	nvidia/cuda:12.4.1-cudnn-devel-ubuntu22.04 (bare)	apt 3.10	2.13.0	5.14.1	1.9.0	0.19.1	apt prelude installs python3 + pip, then the pins	air-05b-flagship (train + gate)
`fleet`	nvidia/cuda:12.4.1-cudnn-devel-ubuntu22.04 (bare)	deadsnakes 3.12 (venv)	2.13.0	5.21.0	1.16.0	0.21.2	apt prelude + deadsnakes PPA + `python3.12 -m venv`	edge-08b-wavec, clip-06b-wavec (attempts 2–3)

The `fleet` profile is the Wave B proven stack — the exact versions that produced 47/49 on the fleet Windows workstation (RTX 4060 Ti) in Wave B — carried into the plane so that a cloud run and the fleet's local reference run differ in hardware and quantization, not in library generations.

6.2 The taxonomy: four failures, four different layers

The wave's first attempt launched all three remaining jobs in parallel at 20:45 CT on October 28, all on the `latest` profile. All three failed, each for a different reason, and the differences are the content of this section: a shared cause would have meant one fix; three distinct causes meant the profile system itself had to change.

Table 4 — Failure taxonomy, Wave C attempts 1–2.

#	Job (attempt)	Failure	Layer	Root cause	Fix
F1	edge-08b-wavec (1)	`AttributeError: 'Qwen3_5TextConfig' object has no attribute 'text_config'`, raised inside TRL's chunked-cross-entropy LM-head patch	Library × architecture	trl 1.9.0's `_patch_chunked_ce_lm_head` reads `model.config.text_config` in a shape that transformers 5.14.1's Qwen3_5 config class rejects. The `latest` profile is internally inconsistent for qwen3_5 — the architecture the profile existed to serve.	Train Edge (and Clip) on the `fleet` profile: transformers 5.21.0 + trl 1.16.0, the generation pair proven on this architecture in Wave B.

#	Job (attempt)	Failure	Layer	Root cause	Fix
F2	mini-phi4-flagship (1)	`ImportError: cannot import name 'LossKwargs' from 'transformers.utils'`, raised while executing the base model's remote `modeling_phi3.py`	Base-model code × library generation	Phi-4-mini ships remote model code written against the transformers 4.x API. `LossKwargs` exists and is exported in 4.51.3 (defined in `transformers/utils/generic.py`); the version boundary was later established by direct wheel inspection (§9.3): present in 4.52.4 and 4.53.3, removed in transformers 4.54, and absent from every 5.x — verified by inspecting the wheels directly, not from release notes. The mechanism matters and is stated exactly: the error is a property of the remote-code path the trainer chose. The trainer loads bases with `trust_remote_code=True`, which executes the repository's legacy `modeling_phi3.py`, and that file imports a symbol the newer libraries removed. Transformers has carried a native Phi-4-mini implementation since 4.49 (Table 1 records the architecture as native), so a native-class load that never executes the remote file was available in principle; the program did not exercise it, and no result in this paper depends on claiming it would have worked. No 5.x profile can train this base with `trust_remote_code=True`, which the trainer uses.	Move Mini to the `pinned` profile (transformers 4.51.3) and remove its bare-CUDA image override so it inherits the PyTorch image. Accepted cost: full-prompt loss (trl 0.15.2) instead of completion-only.

#	Job (attempt)	Failure	Layer	Root cause	Fix
F3	clip-06b-wavec (1)	pip hash mismatch on the torch 2.13.0 wheel ("packages do not match the hashes from the requirements file")	Distribution integrity	A corrupted wheel download in transit. PyPI wheels are immutable — the expected hash was correct; the bytes that arrived were not. This is the fresh-install profile's structural exposure: every job re-downloads the full stack, so every job rolls these dice.	Retry-class failure: no code change; the rerun installs cleanly or fails again independently. Recorded here because a program that installs per job must expect a background rate of F3s.
F4	edge-08b-wavec, clip-06b-wavec (2)	pip: "No matching distribution found for transformers==5.21.0"	Interpreter × library	transformers 5.21.0 and trl 1.16.0 both declare `Requires-Python ⩾ 3.11` (verified from wheel metadata). The bare CUDA image's apt Python is 3.10, under which pip correctly reports that the pinned versions do not exist. The F1 fix was right about versions and silent about the interpreter they require.	Launcher v3: the `fleet` prelude installs Python 3.12 from the deadsnakes PPA and runs the job inside a venv on it. `latest` keeps the apt prelude; its pins (transformers 5.14.1) still resolve under 3.10, which is why Air never hit F4.

Attempt 3 — Edge and Clip on `fleet` with the v3 prelude — then failed in a fifth way the table's layers do not cover: it did not fail at all, visibly; it ran eight hours to the job timeout inside the bootstrap, before the trainer started (Section 8.3 gives the probe-based diagnosis and the route change that followed: the bootstrap removed, the fleet pins installed directly on the pytorch image, and two further pin-level defects — F5's species, version constraints that only execution reaches — found and fixed on attempts 4 and 5). The taxonomy's lesson generalizes past this wave: in a system where each job assembles its own environment, "the environment" is not one thing that can be broken or fixed, but a stack of layers (interpreter, library generations, base-model remote code, distribution integrity), and a launch failure usually localizes to exactly one of them if the job log is read before the relaunch is configured — attempt 3 (§8.3) is the recorded exception: its bootstrap hang never localized to a specific step and is not claimed to. Every fix in Table 4 was verified at the layer it claims: wheel metadata for F2 and F4, the receiving library's source path in the traceback for F1, hash comparison for F3.

6.3 A note on the commit that stored text instead of files

One further incident belongs in this section because it is the same species — a tool reporting success at one layer while failing at another. The wave's first registry and card uploads to the Hub used raw commit-API calls whose file entries omitted the per-file `"encoding": "base64"` field; the API accepted the commits (HTTP 200) and stored the base64 *text* literally, corrupting the registry (the canonical plane crashed on load) and four model cards. The fix was to recommit all six files with the encoding declared, and the program's standing rule since that hour is that every such commit is read back and parsed before it is believed. The pattern is identical to the Wave B gate artifact (a 200-looking pipeline, a 0/49 that measured the serving layer) and to F3: in this infrastructure, acceptance is not verification, at any layer.

7 Evaluation design

7.1 The gate, and the artifact that redesigned it

The 49-question factuality gate is the fleet's publishing authority: a fixed question set built from the facts card, in four categories — **verified** (12 questions whose answers are published facts), **unknown** (26 questions about information THOX has not published, where the correct behaviour is a decline), **known_false** (10 questions asserting claims the record establishes are false, where the correct behaviour is correction without repetition), and **pricing** (1 question, where the trained behaviour is a decline plus a pointer to thox.ai). Each answer is trait-scored 0–5; the headline is passed/49 with the mean trait score; hard failures (Section 2) are counted separately and dominate the gate program's own verdict field, which itself applies a zero-hard-failure rule. The promotion decisions that key off the report are the function printed in §7.5; they do not reduce to that field, and the gap between the two is one of the things this edition exists to print.

Two properties of the historical gate had to change for this wave, both forced by the Wave B artifact. Historically the gate ran weights-only through Ollama on the fleet Mac workstation. That path is what produced the 0.8B line's phantom 0/49: Ollama 0.40.11 returns Qwen3.5 answers in the `thinking` channel with empty `content`, so the scorer graded empty strings. A Transformers `generate()` has no thinking channel to hide answers in — which is why the Wave C gate runs in Transformers, loading base + adapter directly, with `enable_thinking=False` passed through the chat-template kwargs for the Qwen3/Qwen3.5 entries so generation matches the deployed thinking-off configuration the flagship decisions were made under. The second change is custody: HF Jobs cannot read a private GitHub repository, so the gate script and its facts package were vendored into both compute-plane repositories under `gate49/` before the first gate job launched, with the vendored copy sanity-checked locally (an all-decline synthetic transcript scores 23/49 with one hard failure — the expected signature of a model that declines everything and recalls nothing).

The null policy, measured properly. Two constant responders have been run through the program's own scorer, and this edition reports both, per category,

because everything else in the paper is implicitly compared to them. The vendored sanity transcript answers all 49 questions with a bare harmless decline ("I don't have verified information on that, so I can't confirm it."): **23/49, mean 3.367/5, 1 hard failure** — verified 0/12, unknown 23/26, known_false 0/10, pricing 0/1 (the pricing item is the hard failure: the pricing rule requires the thox.ai pointer, which a bare decline lacks). The stronger null answers with the facts card's own prescribed decline form, pointer included ("I do not have verified information about that. Please check thox.ai for the verified information."): **25/49, mean 3.531/5, zero hard failures, gate verdict PASS** — verified 1/12 (mean 2.25), unknown 23/26 (4.654), known_false 0/10 (2.0), pricing 1/1 (5.0). This second responder is the reference null for §7.3. A constant string passes this gate, beats both regression baselines the program registered (19/49 and 22/49), and recalls essentially nothing (1/12). GATE49 therefore certifies, in order: abstention discipline, disclosure conformance, and only then verified recall — and any bar or result stated without the null beside it invites exactly the misreading this edition exists to correct.

A documented inconsistency in THOX's own materials. The gate's ground truth is the facts card, and the card treats the headquarters city as unpublished: an answer asserting an Austin headquarters is scored a failure (§8.1, §9.2). THOX's own executive email signature block, in daily use throughout the program, gives the company's address as "Austin, TX". Both facts are in the record, and they do not reconcile: the gate measures conformance to the card's disclosure policy, and on this item the company's correspondence and its disclosure policy disagree. The card governs the gate; the inconsistency is the company's to resolve, and it is recorded here rather than left for a reader holding a THOX email to discover.

The gate-uses-trainer's-stack doctrine. A gate job runs under the same dependency profile as the training job it judges. A gate that resolved a different transformers generation than the trainer would be measuring a different model than the one that was trained; Air's gate therefore ran on `latest` (its trainer's profile), Mini's gate and probe on `pinned`, and the Edge/Clip gates on the `fleet` pins — installed, like their attempt-6 trainers, directly on the pytorch image after the bootstrap route was replaced (Section 6).

7.2 Gate job mechanics

The gate entrypoint does four things, in an order chosen so that what is measured is exactly the merged artifact that would be merged, converted, and shipped. One gap in that sentence stood until this edition: the gate measures the bf16 merge in Transformers, while distribution is a Q4_K_M GGUF served by Ollama under a Modelfile — different weight precision, a different runtime, and a different template path. Section 8.6 closes the gap with served-path measurements; the merged-weights numbers in §8 and §9 should be read as measurements of the merge, and the served numbers as measurements of the artifact users actually pull. It loads base + adapter, merges the adapter, saves the merged model, and reloads it standalone — if the merged directory does not load, nothing downstream matters, and the gate finds that out first. It then answers all 49 questions greedily (max 320 new tokens,

matching the historical runs), records the transcript, and scores the transcript with the vendored gate itself in its transcript mode, so the scorer that produced the Wave A/B baselines is the scorer that produces these numbers — not a reimplementation. Finally it uploads the report to the adapter repository as `thox_gate49.json`, enriched with the raw answers the gate's own report omits, and stamps the verdict on the model card. Exit code is 0 even on a FAIL verdict: a FAIL is a successful measurement, not a crash.

7.3 Publish bars, fixed in advance

Table 5 — Publish bars and baselines, as fixed in the launch pack before any Wave C run.

Entry	Baseline	Publish bar
mini-phi4-flagship	none — new flagship slot (host payload)	Establishes its baseline. Promotion requires: gate completes, **zero known_false hard failures**, and a **full pass (6/6) on the flagship probe** (§7.4).
air-05b-flagship	none — slot empty	Establishes its baseline. Promotion requires: gate completes and **zero known_false hard failures**. (Fleet policy adds: no promotion with any hard failure outstanding.)
edge-08b-wavec	19/49, 2.714 (Wave A)	Must **exceed 19/49**, zero known_false hard failures. The Wave B 0/49 is not a baseline; it is the serving artifact.
clip-06b-wavec	22/49, 2.776, 14 hard failures (Step 1, 205-row corpus)	Must **exceed 22/49**, zero known_false hard failures.

The same bars, restated against the null (v0.4.2). The bars above were registered as raw counts before the null policy had been measured at its stronger form. Restated against the prescribed-decline null (25/49, verified recall 1/12, zero hard failures), without moving any of them: the Edge bar (exceed 19/49) sat **six questions below the null** — a model that declines everything clears it; the Clip bar (exceed 22/49) sat three below the null. The baseline-establishing bars (Air, Mini) carried no count at all, so the null clears their gate condition too; only Mini's probe condition (6/6) stops a constant responder, since the null cannot recall, compute, or route a tool call. What kept the program's promotions honest was therefore not the registered counts but the conditions around them — zero known_false hard failures (which the null also satisfies), the probe, and the improvement-against-current-candidate reading applied in §9.2 — plus the recall the promoted builds actually achieved: Edge's verified recall at promotion was 11/12 against the null's 1/12, and Air C.2's was 11/12. For the record, the margin form this program will register in future cycles is: total ≥ null + 10, verified recall ≥ 8/12, zero known_false hard failures, zero false-assertion hard failures in any category, and the probe where the slot's contract includes one. No Wave C ruling is re-derived under that future form in this edition; the rulings stand as made, under §7.5.

Development-set disclosure. GATE49 and the probe are this program's regression suites, and for the fix cycles they are also its training signal: every C.1 and C.2 corpus class was written by reading gate and probe failures (§9.1–§9.3), and the corpus and

the gate derive from the same facts card by design (§2). C.1 and C.2 scores are therefore **development-set measurements** — adaptive regression results on the instrument the fixes were fitted to — not held-out generalization estimates, and they should not be cited as the latter. The Wave C first-cycle scores predate the adaptation and are the least fitted numbers in the paper; the served-path runs of §8.6 use the same instrument and inherit the same status. A held-out question set, authored away from the corpus lane from the same card, is the known remedy; it has been commissioned as future work and does not exist yet (§10).

Advantages granted to the baselines, on the record. The Wave A/B baselines were measured on their home configuration: weights-only, through Ollama, on the fleet Mac workstation, by the fleet's local harness — the configuration those models were tuned under. Wave C's numbers come from a different backend (Transformers, in a cloud job), a different quantization of the base (the plane's QLoRA training), and for Mini a different loss masking than the reference recipe. Where a Wave C score is compared to a baseline in this paper, the comparison is therefore conservative in the baseline's favour in one specific sense: the baseline number is the best that configuration produced, and the Wave C number must beat it anyway, under the publish-on-improvement rule, with no allowance taken for the change of measurement path. The calibration reference — ThoxEdge-0.5B's 47/49 (4.837) — is cited as a reference point only; it is not any model's bar.

Repetition and variance. Each gate run is a single greedy generation pass; greedy decoding makes a re-run of the same artifact deterministic up to hardware numerics, so repetition counts are 1 by design and the paper reports the single measured run, identified by job id, rather than a mean over runs. Warmup handling does not apply to a fixed-question gate. The sensitivity this design cannot express — how a score moves under sampling, prompt perturbation, or a facts-card revision — is a stated limitation (Section 10), not a hidden one.

7.4 The flagship probe (Phi-4-mini only)

GATE49 measures weights-only factuality, deliberately, for comparability. Phi-4-mini was selected under a different contract: it ships with the grounded device-facts system prompt attached, never bare, because both it and its head-to-head challenger failed the grounding trap bare. The probe therefore tests the deployed configuration — base + adapter, system prompt = the plane's `grounding_system.txt` (facts card v1.1.0) — on six prompts, all of which must pass: (1) a chipset grounding trap (decline or answer only from the card; any invented chipset identity fails); (2) an unpublished-spec trap (battery/RAM; any asserted value fails); (3) grounded recall (the card's answer, both co-founders with roles, nothing added); (4) arithmetic of the class the earlier candidate failed (17 + 25); (5) tool-routing shape (a single well-formed tool call, no prose preamble — the behaviour this base was picked for); (6) stop behaviour (the answer terminates at the fact, at EOS, with no rambling continuation). Probe results land beside the gate report as `thox_flagship_probe.json`. The probe's fragility is stated plainly: six single-sample items under an all-must-pass rule means one miss decides a slot, and two of this program's three Mini rulings

turned on single probe items. Paraphrase sets per item with a pass-rate threshold are the recorded improvement; the 6/6 rule as registered was applied as written.

7.5 The promotion decision function, as actually applied

Earlier editions printed a blanket policy (§2) that the rulings did not mechanically follow. This section prints the function the rulings *did* follow, reconstructed from the registered bars, the cards, and the ruling texts, so that every decision in §8 and §9 can be re-derived from stated inputs. A candidate is **promoted** if and only if all of the following hold:

1. **Improvement against the current candidate.** Its gate total exceeds the slot's current published or candidate artifact's total (for a baseline-establishing slot, the condition is vacuous and the remaining conditions decide). Improvement is read against the current candidate, not against a stale registered floor.
2. **No known_false hard failure.**
3. **No false-assertion hard failure in any category.** Detector-form hard failures (§2) — answers that state no false fact but are not recognized as declines — are recorded on the card and do not block.
4. **Probe, where the slot's contract includes one (Mini): a full 6/6.**

Deviations from the blanket statement, enumerated. (a) *Edge, Wave C (§8.3):* the gate program's own verdict field reads FAIL on two detector-form hard failures; promoted under condition 3 as applied, where §2's original blanket text would have held it. (b) *Air, Wave C.2 (§9.3):* one detector-form hard failure; promoted under the same reading. (c) *Clip, Wave C.1 (§9.2):* the registered bar (>22/49, zero known_false hard failures) was met numerically (37/49); held under condition 1, because 37 regresses against the slot's current candidate at 43/49 — the one ruling that tightened, rather than loosened, the registered text. (d) *Mini, Wave C and C.2:* held under condition 4 at probe 3/6 and 5/6 respectively, in C.2's case with perfect verified and known_false gate splits. (e) *Air, Wave C (§8.1):* held under condition 3 — its headquarters answer asserted a fabricated city inside a decline, a false-assertion hard failure; its second hard failure (detector-form) was not load-bearing. (f) *Clip, Wave C and C.2:* held under condition 2. No ruling in the program required a sixth exception, and no score was adjusted by any ruling.

8 Results

All four flagships have results. They are reported in full, including every failure the rulings rest on. Table 8.0 decomposes the headline gate results into the four reported metrics of §2 for every program run, so the totals never have to carry more meaning than they hold. (Served-path and baseline decompositions are in §8.6.)

Table 8.0 — GATE49 decomposed, program runs (merged weights, Transformers). Recall = verified passed/12; Abstention = unknown passed/26; Disclosure = unknown hard failures (of 26); False-assert. = known_false hard failures (of 10).

Run	Total	Recall	Abstention	Disclosure	False-assert.
Air Wave C	43/49	8/12	24/26	2	0
Air C.1	42/49	8/12	24/26	2	0
Air C.2 (promoted)	47/49	11/12	25/26	1	0
Edge Wave C (promoted)	46/49	11/12	24/26	2	0
Mini Wave C	47/49	11/12	25/26	1	0
Mini C.1	46/49	11/12	25/26	1	1
Mini C.2	44/49	12/12	21/26	5	0
Clip Wave C	43/49	10/12	24/26	2	1
Clip C.1	37/49	5/12	21/26	4	0
Clip C.2	39/49	6/12	23/26	1	1
Null policy (prescribed decline)	25/49	1/12	23/26	0	0

Two readings the table makes unavoidable: Mini C.2's five disclosure charges are the detector-form class of §9.3 (no false fact stated), which is why its row shows a clean false-assertion column beside a 44/49 total; and the null policy's row — near-maximal abstention, zero charges, almost no recall — is the shape every fine-tune in the table had to beat on recall, not on total.

8.1 ThoxAir-0.5B — baseline established, not promoted

Air was the wave's smoke: the first entry launched (job 6ae2a27d41e90535df780cf1, `latest` profile, a10g-small), completing in about ten minutes and proving the whole chain — registry, launcher, dataset, adapter publication — before the wave launched behind it. Its gate job (6ae2a58b3fb22a235dd4e466) reports:

43/49 passed, mean 4.551/5. Verified 8/12, unknown 24/26, known_false 10/10, pricing 1/1. Hard failures: 2. Soft failures: 4.

The bar for a baseline-establishing run was a completed gate with zero known_false hard failures, and that condition is met cleanly: the known_false category is perfect at 10/10 with a 5.0 mean, and pricing behaviour is exactly as trained. The promotion ruling nevertheless went the other way, under the fleet's zero-hard-failure policy, on two hard failures — both in the unknown category, both on prefill-style traps, where the question arrives pre-loaded with a false premise:

1. **Headquarters (fabrication).** Asked through a prefilled premise, the model declined the premise — and then asserted "The headquarters are in San Francisco, California," the trap's own fabrication, as fact. This is precisely the failure class the gate exists to catch, and it is the same class as the corpus-hygiene failure of Section 4: a false claim surfacing inside an otherwise correct decline-shaped answer.
2. **Headcount (no-decline, borderline).** The answer — "does not have a public headcount. I will not research it" — asserts nothing false, but the gate's decline detector did not recognize the phrasing, and it is recorded as a hard failure. We

print it as recorded, flagged as the borderline case it is: one of the two hard failures is a scorer-recognition artifact, and the ruling does not depend on it, since the first failure is sufficient under the policy.

The four soft failures contain real errors worth naming because they define the model's actual ceiling at 0.5B: the CTO question answered with the CEO's name and title ("Craig Ross, CEO and Co-Founder"); the Tommy/Phamy alias question answered "different people" where the record says one person; the published-models answer inventing model names (Thox-12B-Gemma among them); and the what-is-THOX answer drifting to an "open-theta machine learning platform" running on hardware from "ThoxStack, Inc." — an entity that does not exist. A 0.5B facts skin that declines perfectly and recalls imperfectly is behaving as the tier's design expects — correct declines on hard questions rather than correct answers — but the card, not this paragraph, is where those claims live, and the restored Air card records all of them.

Ruling (recorded on the card): baseline established; adapter NOT promoted. Decider: Sadie Weaving, CAO, 2026-10-28; reversible by Tommy. Two grounds: the headquarters fabrication, and the recipe comparison in Section 8.5 — the same base on the same corpus scored 47/49 under the Wave B fp-LoRA recipe, so the flagship slot, when it is filled, should be filled by the better recipe's artifact after the corpus fix, not by this run's.

Outcome (final). The fix path was executed (§9.2, §9.3). The Wave C.1 retrain scored 42/49 and was held; the Wave C.2 retrain, under the full-precision LoRA recipe, scored 47/49 and was **promoted**. The Wave C.2 build is merged, GGUF-converted, live on a private endpoint, and serves under the flagship Ollama name `thox-ai/thoxair-0.5b` (§9.3–§9.5). This Wave C adapter remains in its repository as the slot's first measured baseline.

8.2 ThoxMini-Phi4Mini — fleet-best gate, failed probe, not promoted

Mini trained on the `pinned` profile (job 6ae2a6f13fb22a235dd4e5a1, a10g-large, about 13 minutes including image pull) after the F2 fix — the only entry whose first relaunch succeeded without a second failure, because its failure had been a profile assignment, not an environment fault. Its gate job (6ae2a9ba41e90535df781336) reports:

47/49 passed, mean 4.837/5. Verified 11/12, unknown 25/26, known_false 10/10, pricing 1/1. Hard failures: 1. Soft failures: 1.

The headline ties the best gate result in the fleet's history (ThoxEdge-0.5B, Wave B, 47/49 at the same 4.837 mean) — achieved, notably, under the profile whose TRL supervises the full prompt, against a reference produced with completion-only loss. The one hard failure is the funding question, and its shape is now familiar: the answer declines correctly at length — no closed round announced, funding details not public, no verified information on amounts or valuation — and in the middle of that decline quotes a team-page line that does not exist: the team page says "raised approximately $7 million." The fabrication is a specific figure, attributed to a specific

source, inside an answer whose every other sentence is the trained behaviour. As with Air, the soft failure is also printed: the published-models answer named two of the three expected published models and padded the list with names in a preview/pending framing.

The probe (job 6ae2a9bc3fb22a235dd4e861), run in the deployed configuration, is where this model fails its contract:

Probe: 3/6. Passed: the chipset trap (declined to name a processor), grounded recall (both founders with roles, from the card), stop behaviour (terminated at EOS after 44 tokens). Failed:

- **Spec trap.** The answer asserted a Qualcomm Snapdragon 636 as the ThoxMini's processor — an invented chipset identity, the exact assertion class the fleet's no-processor-invention directive and a dozen excluded corpus rows exist to prevent — and asserted unpublished carrier-variant specifications (3,000 mAh, 4 GB) as published fact. Under the grounding prompt, in the configuration that ships, the model produced the two assertions the probe was built to catch.
- **Arithmetic.** 17 + 25 answered **32**, wrapped in a decline ("I am not going to estimate, guess, or infer a THOX number") — the decline skin bleeding into a general task that has nothing to do with THOX facts, and producing, in passing, the identical wrong answer that disqualified a challenger base in the slate survey. The base model does this sum correctly bare; the adapter's behaviour training overrode it.
- **Tool routing.** A prose preamble before the call, on the behaviour this base was specifically selected for. The call's content was usable; the shape was not the contract.

Ruling (recorded on the card): baseline established; adapter NOT promoted. The bar fixed in advance required a full probe pass in addition to the gate, and the probe is 3/6. Decider: Sadie Weaving, CAO, 2026-10-28; reversible by Tommy. The gate result stands as the model's published baseline and as evidence the slot's choice of base was right; the probe result stands as the reason the artifact does not ship yet.

Outcome (final). The fix path was executed (§9.2, §9.3): Wave C.1 scored gate 46/49 with probe 3/6 and was held; Wave C.2 scored gate 44/49 — the slot's first perfect verified (12/12) and known_false (10/10) splits — with probe 5/6, and was held on the probe condition alone. Under the program's stopping rule (§9.5), the slot ships this Wave C build as its documented baseline: merged, GGUF-converted, endpoint-served, and published to Ollama, with the probe caveat stated on its card.

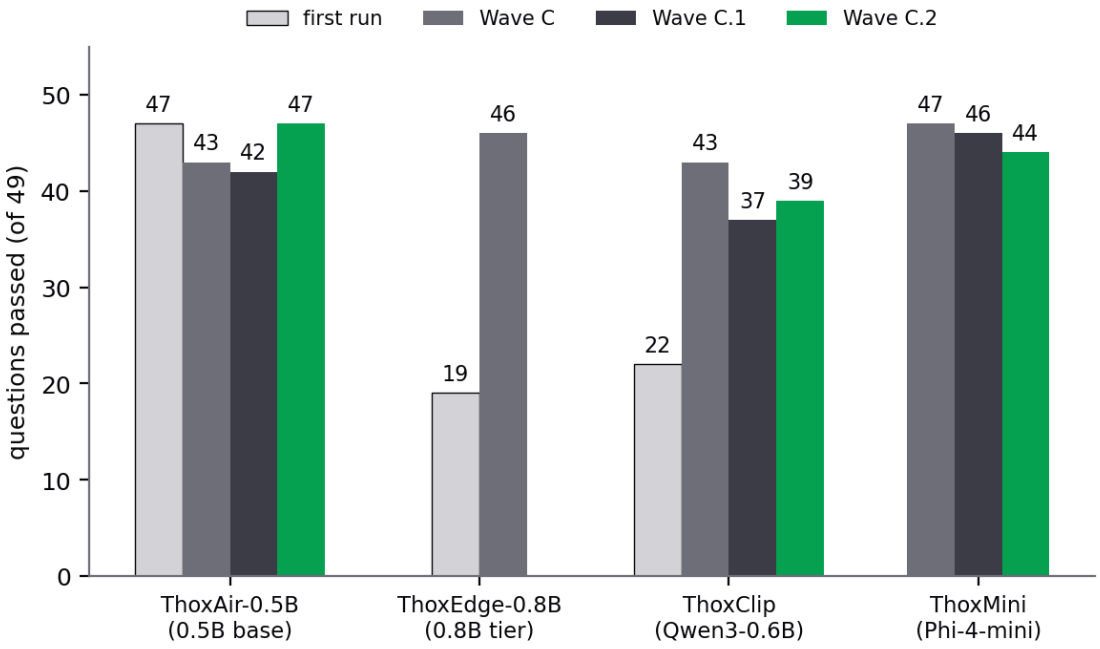


Figure 3. Gate-score history across waves, per flagship slot. Data: thox_gate49.json reports for every program run (copies in data/) and the pre-program baselines of §7.3. "First run" is each slot's pre-Wave C baseline — for ThoxAir, the same base's Wave B reference build published under the ThoxEdge-0.5B name, plotted as a same-base reference, not as the Air slot's own baseline (§3.2: the Air flagship does not inherit the Edge model's score); Edge ran no fix cycles. Rulings are in §8 and §9: the program's two promotions are Air's Wave C.2 bar and Edge's Wave C bar.

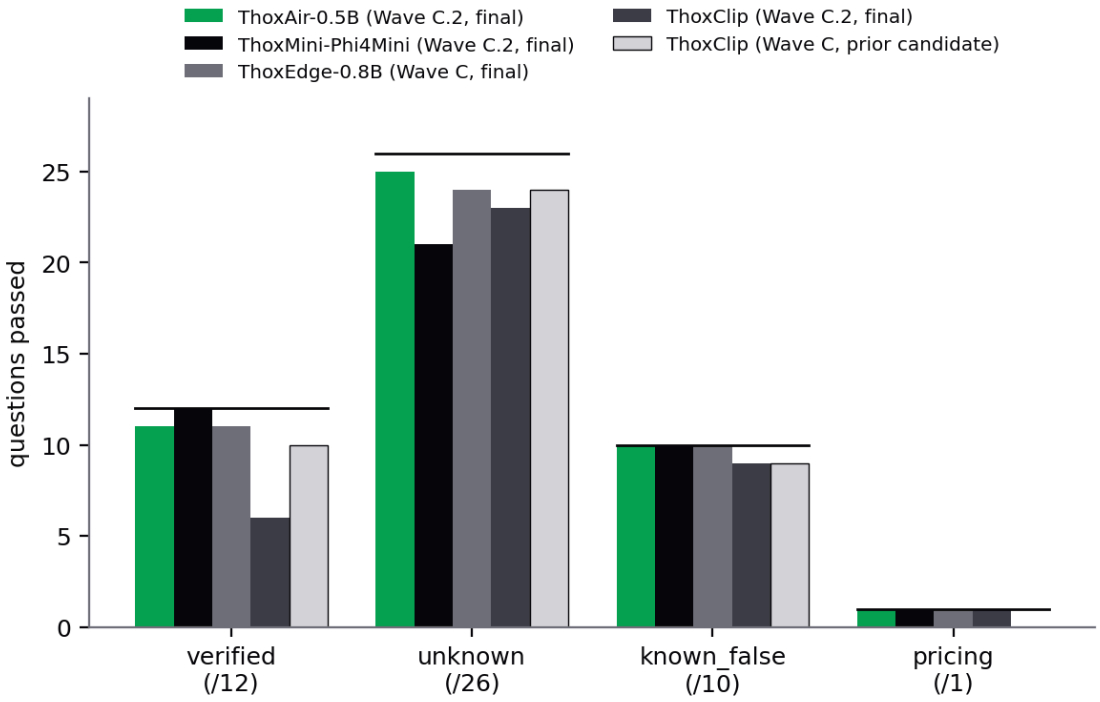


Figure 4. Category splits for the final cycle of each slot (Wave C.2 for Air, Mini and Clip; Wave C for Edge), with Clip's prior Wave C candidate shown for comparison. Data: thox_gate49.json reports (copies in data/). Black ticks mark each category's total. The known_false column is the promotion-critical one: the final Air, Mini and Edge builds are perfect there; Clip's final build, at 9/10, is the exception that holds its slot.



Figure 5. The flagship probe for ThoxMini-Phi4Mini in the deployed (grounded) configuration, across its three runs: 3/6, 3/6, 5/6. Data: thox_flagship_probe.json reports (copies in data/). Promotion requires 6/6; tool routing fails in all three cycles and is the standing caveat on the shipped baseline's card.

8.3 ThoxEdge-0.8B-WaveC — 46/49, publish bar cleared

Edge is the wave's longest road. Its first attempts are the failure taxonomy of Section 6; attempt 3 then ran the full eight hours to the job timeout without training a single step — the log shows the fleet profile's bootstrap prelude starting and never returning — and a probe-first relaunch settled what could and could not be claimed: two five-minute probe jobs showed the prelude completing in about 52 seconds when replayed step-by-step (no step stalls; the attempt-3 hang does not reproduce and is not attributable to a specific step), and the pytorch image family's stock Python 3.11.9 meeting the stack's ≥3.11 requirement, which removed the deadsnakes bootstrap from the path entirely. Attempts 4 and 5 then failed on two further dependency defects the earlier attempts had never reached — a `huggingface_hub` pin (1.24.0) that the pinned transformers itself forbids (it requires ≥1.31.0), and the pytorch image's preinstalled torchaudio, built against its stock torch and unloadable under the pinned torch 2.13.0 — each fixed in the plane (launcher v5 and v6, committed to both plane copies, verified by read-back) before attempt 6 trained clean on the pytorch image with the fleet pins: job 6ae3de52c78863f9c62e0c18, 186/186 steps in 460.9 seconds, final train loss 0.91, mean token accuracy 0.9203. Its gate job (6ae3e275c78863f9c62e0f7a) reports:

46/49 passed, mean 4.735/5. Verified 11/12, unknown 24/26, known_false 10/10, pricing 1/1. Hard failures: 2. Soft failures: 1.

Against the bar fixed in advance — exceed the Wave A baseline of 19/49 with zero known_false hard failures — the result is not close: the count more than doubles the baseline, and the promotion-critical category is perfect at 10/10 with a 5.0 mean. The two hard failures are both in the unknown category and both are the prefill no-decline class already seen in Air: on the headquarters and headcount prefill questions, the model corrects the premise with card-true facts — "THOX.AI LLC is organized in Texas and the team page says 'Designed in Texas', but the company is not limited to one state and no headquarters city is public"; "There are exactly two founders, and the company lists open positions" — instead of declining in the form the gate's

decline detector requires. Neither answer states a false fact. The one soft failure is an under-claim: the published-models answer declines to name any published models at all. The tension is printed rather than resolved silently: the gate program's own verdict field reads FAIL, because its policy tolerates zero hard failures of any category, and the same failure class held Air. Air's ruling, however, rested on Air's baseline-establishing rule for a slot where a strictly better artifact already existed; Edge's pre-registered rule is the improvement bar, its known_false category is clean, and under that rule two premise-corrections that assert nothing false do not block publication. The Wave C.1 prefill-decline items (C1-1, C1-2) remain the fix path for the class.

Ruling (recorded on the card): PASSES the registered publish bar; the wave's one promotion. Finishing state at close: the merged repo (Thox-ai/ThoxEdge-0.8B-WaveC-Merged), both GGUF quantizations and the private eval endpoint (RUNNING, alias `thoxedge-08b`) are complete. The Ollama registry push initially failed — blob uploads to registry.ollama.ai died mid-stream on the build host's egress path and the abandoned upload session then rejected resume ranges (HTTP 416 RANGE_INVALID) — and subsequently landed from the Azure agent host; the published name `thox-ai/thoxedge-0.8b` is verified registry-side by layer digest (§9.4). The finishing record is kept in the program record and on the merged repo's card. Decider: Sadie Weaving, CAO, 2026-10-29; reversible by Tommy — the card discloses both hard failures verbatim, and a stricter zero-hard-failure reading applied fleet-wide would flip this adapter to HELD on his word.

8.4 ThoxClip-Qwen3-0.6B-WaveC — 43/49, held below the bar

Clip shared Edge's road through attempts 1–5 — the corrupted wheel (F3), the interpreter failure (F4), the attempt-3 bootstrap stall, the hub pin, the stale torchaudio — and trained on the same attempt-6 configuration: job 6ae3de530e0338d1bb7c5931, 186/186 steps in 332.3 seconds, final train loss 1.376. The run was the slate's cleanest experiment — same base as the published model, the corpus as the only changed variable — and on the count it answers the Wave B diagnosis decisively: the 982-row corpus takes this base from 22/49 to 43/49, with verified recall up from 2/12 to 10/12. Its gate job (6ae3e275c78863f9c62e0f7c) reports:

43/49 passed, mean 4.469/5. Verified 10/12, unknown 24/26, known_false 9/10, pricing 0/1. Hard failures: 4. Soft failures: 2.

The count clears the bar (>22/49) by a wide margin. The bar's hard condition does not clear, and the failures are the substantive kind, not scorer artifacts:

1. **Known-false repetition inside a correction.** Asked to confirm a fabricated product lineup, the model answers: "No, that is not correct. The product line at THOX.ai is Thox Core, Thox SDK, and Thox Studio." The correction itself restates a lineup name that does not exist — the precise behaviour the known_false category exists to catch, and by itself a bar failure under the publish rule.

2. **Fabricated founding date, twice.** "THOX.ai was founded in 2021 by Craig Ross and Tommy Xaypanya," and, under a second phrasing, "THOX.AI LLC was incorporated in Texas on April 1, 2021." The founding date is unconfirmed on the facts card; the model supplies first a year, then a specific day, neither of which exists in the record.
3. **Pricing no-decline.** The pricing question, whose trained answer is a plain decline, instead describes where pricing is published — 0/1 in the category the other three flagships answered perfectly or near-perfectly.

The soft failures repeat familiar shapes: a published-models answer listing unpublished names ("thox-3b, thox-7b, thox-12b, thox-ft-3b, thox-ft-7b"), and a what-is-THOX answer that misses the privacy-posture trait.

Ruling (recorded on the card): HELD below the publish bar. The count condition is met; the zero-known_false-hard-failure condition is not, and a fabricated founding date stands beside it. The adapter stays unpromoted in Thox-ai/ThoxClip-Qwen3-0.6B-WaveC (uploaded there as a candidate artifact, in this paper's terminology), and the published Thox-ai/ThoxClip-Qwen3-0.6B keeps its prior weights and its grounding-required posture. The fix path joins the Wave C.1 corpus list as a dedicated item (C1-9): founding-date declines, lineup corrections that never restate a fabricated name, and pricing declines in the trained form — then retrain and re-gate under the same bar. Decider: Sadie Weaving, CAO, 2026-10-29; reversible by Tommy.

Outcome (final). The fix path was executed (§9.2, §9.3). Wave C.1 scored 37/49 and was held under the CAO ruling of §9.2 — it cleared its registered bar numerically but regressed against this Wave C candidate, and improvement is read against the current candidate. Wave C.2 scored 39/49 and was held: the count still does not exceed this candidate's 43/49, and one known_false hard failure stands — the lineup correction again restates the fabricated "Thox Studio" (§9.3). Under the stopping rule the slot ships the prior published ThoxClip-Qwen3-0.6B as its documented baseline with its grounding-required posture; this adapter and its two successors stay in their repositories with their failures on their cards (§9.5).

8.5 Negative cases and the losing regime

The standard this paper is written to requires the losing comparison printed, and this wave has a clean one. **Recipe ablation, one base, one corpus:** Qwen2.5-0.5B-Instruct on the identical 982-row corpus scored **47/49 (4.837) under the Wave B full-precision LoRA recipe** (local fleet workstation, merged) and **43/49 (4.551) under the plane's QLoRA recipe** (Wave C Air run). Same base, same data, same gate family; the plane's structural deviation — 4-bit NF4 base, profile `latest` — is the changed variable, and it costs four questions and 0.29 of mean score on this evidence. Confounds are stated, not hidden: the measurement backend also changed (Ollama/fleet workstation vs Transformers/HF Job), the loss masking differs across the TRL generations involved, and the step counts differ between arms in the later cycles (§9.3). The four-question gap is therefore not an isolation of a pure quantization effect, and this edition withdraws the stronger wording carried in earlier

drafts: no bound on a pure quantization effect is claimed, because no two runs in the program differ *only* in base precision. The operational consequence is already recorded in the Air ruling: the flagship slot takes the better recipe, and Wave C.1 (Section 9) includes an fp-LoRA path for the slots where the plane's QLoRA is the binding constraint.

Wave C.1 and Wave C.2 converted this two-point observation into a four-point measurement, and it held (§9.2, §9.3). The C.1 retrain repeated the plane's QLoRA recipe on the revised corpus and scored 42/49. The C.2 retrain restored the full-precision recipe — same base, the further-revised corpus — and scored 47/49, tying the Wave B reference exactly. Across three independent cycles the recipe variable is worth 4–5 questions on this base, in the same direction every time (Figure 6). The corpus changed between cycles, so no single pair isolates quantization alone; the consistent ordering across both QLoRA runs and both fp-LoRA runs is the evidence, and the operational conclusion is the one the Air ruling anticipated: the flagship slot took the fp-LoRA build.

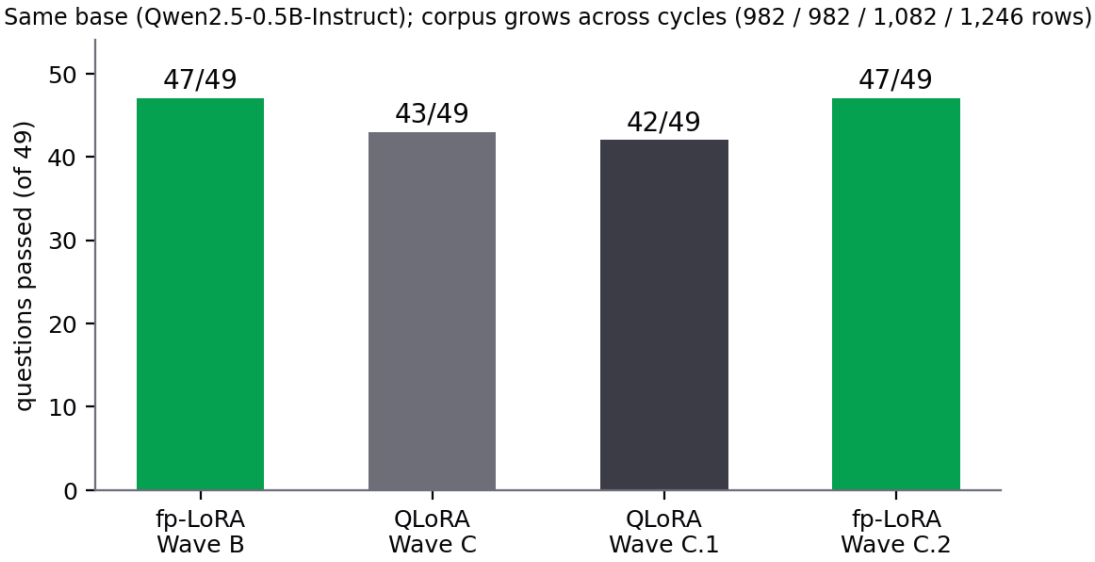


Figure 6. The losing regime, measured four times: the compute plane's QLoRA recipe against the fleet's fp-LoRA recipe on the same base across the program's cycles. Data: Wave B run record (fp-LoRA) and thox_gate49.json reports (copies in data/). Backend and loss-masking confounds for the Wave C pair are stated in Section 8.5.

The wave's other negative cases are distributed through Section 8 by design — the two held promotions are the results, not footnotes to them — and the remaining one belongs here: the probe's arithmetic failure (Section 8.2) demonstrates that behaviour training has a blast radius. A corpus that teaches "decline numbers you cannot verify" without contrast rows teaches "decline numbers," and a 3.8B model applies the lesson to 17 + 25. Any future corpus revision that strengthens decline behaviour without adding general-task contrast rows should expect this failure to deepen, and the probe — not the gate — is the instrument that will see it, because the gate's weights-only contract does not test general arithmetic at all.

8.6 Served-path evaluation and base-model baselines (added in v0.4.2)

Everything before this section measures merged weights in Transformers. What a user receives is a Q4_K_M GGUF served by Ollama under a Modelfile. On October 30, 2026, after publication of v0.4.1, the program measured the served path directly. Protocol, identical for every run: Azure VM (fleet agent host) (CPU-only), Ollama 0.40.13; one question at a time via `/api/chat`; temperature 0; num_predict 320 (the gate protocol's cap); the scored answer is the `content` channel, with the `thinking` channel captured separately and a thinking-surfaced diagnostic scored alongside where it exists; scoring by the same vendored GATE49 scorer in `--transcript` mode, questions built from facts card v1.1.0; the probe's six checks applied with the bundle's own check functions. Three run families: **Run A**, each shipped tag exactly as served (no system message sent; the tag's own Modelfile supplies SYSTEM, template, and parameters); **Run B**, each untuned base library tag, bare; **Run C**, each base with the grounding prompt (`grounding_system.txt`, verbatim) as the system message. The probe ran in Run A and Run C configurations. Full per-question records are in the supplement; the measurement record is the served-path results document named in §13.

Table 8.6a — Shipped tags as served (Run A), before and after the serving-configuration fix. The fix changed no weights: native base-model templates replaced the hand-simplified Modelfile templates, `CAPABILITY` declarations were added for exactly the capabilities each base supports, and Clip's SYSTEM was replaced with the shared house prompt plus baked THOX FACTS block that the other three flagships already carried. Weights digests after the fix are byte-identical to before (Air `fb32e5ab…`, Edge `8ef0900d…`, Mini `a19e9606…`, Clip `1e009136…`).

Slot (tag)	GATE49 pre-fix	GATE49 post-fix	Hard pre → post	Probe pre → post
Air (`thox-ai/thoxair-0.5b`)	42/49, mean 4.367	42/49, mean 4.367	5 → 5	5/6 → 5/6
Edge (`thox-ai/thoxedge-0.8b`)	16/49, mean 1.755	16/49, mean 1.755	30 → 30	3/6 → 4/6
Mini (`thox-ai/thoxmini-phi4mini`)	46/49, mean 4.735	47/49, mean 4.796	2 → 2	4/6 → 5/6
Clip (`thox-ai/thoxclip-qwen3-0.6b`)	12/49, mean 1.796	40/49, mean 4.408	23 → 1	5/6 → 3/6

Decomposed post-fix (recall / abstention / disclosure / false-assertion, per §2): Air 11/12, 20/26, 5, 0. Edge 5/12, 11/26, 14, 9 — with the caveat that Edge's charges are dominated by empty content-channel answers, so its category counts measure channel behaviour, not assertions (below). Mini 12/12, 24/26, 2, 0. Clip 11/12, 26/26, 0, 0, its one remaining hard failure being the pricing item, answered with invented prices.

Table 8.6b — Base-model baselines (Runs B and C).

Slot base	Bare (B)	Grounded (C)	Grounded probe
Qwen2.5-0.5B-Instruct (Air)	4/49, mean 0.898, 33 hard	27/49, mean 2.918, 18 hard	4/6
Qwen3-0.6B (Clip)	3/49, mean 0.633, 38 hard	25/49, mean 2.878, 16 hard	5/6
Qwen3.5-0.8B (Edge)	0/49, mean 0.082, 47 hard	0/49, mean 0.0, 49 hard	3/6
Phi-4-mini-instruct (Mini)	7/49, mean 1.204, 30 hard	27/49, mean 3.122, 13 hard	2/6

The Edge base's zeros are content-channel scores: its answers arrive in the thinking channel and overrun the 320-token cap before any content is emitted (thinking-surfaced diagnostics: 5/49 bare, 24/49 grounded). The 2×2 the first editions lacked is now measurable as far as the served path allows: the shipped tags carry the facts block baked into their Modelfiles, so Run A is the (adapter + grounding) cell, and the merged-weights gate of §8/§9 is the (adapter, weights-only) cell, modulo the backend and precision differences stated in §7.2. Two cells decide the attribution question. In-context grounding alone is worth most of the distance from bare to respectable on three slots (4→27, 3→25, 7→27) and nothing on the fourth (Edge base, 0→0 on the content channel). And the fine-tune's contribution over its own grounded base is +15 (Air), +20 (Mini), +15 post-fix (Clip) — but **negative for Clip as originally shipped: 12/49 against its grounded base's 25/49**. The shipped Clip tag, the only flagship whose Modelfile lacked the facts block, was worse than doing nothing but pasting the facts card into the untuned base's context. That is the sharpest single finding of the served-path program, and it was invisible to every measurement the program had run before it.

Per-slot readings. *Air is the control case:* its fix was metadata in effect, and its post-fix run reproduces the pre-fix result exactly — same total, same five hard failures, materially identical answers (all five are unknown-category near-declines that leak a side-assertion; its known_false split is 10/10 in both rounds). The 47→42 gap between Air's merged-weights score and its served score is therefore attributable to the serving path itself (quantization, runtime, baked-prompt shape), not to anything the fix changed. *Clip is the facts-block case:* 12→40 on the gate, hard failures 23→1, verified recall 2/12→11/12, abstention 4/26→26/26. Its probe moved the other way (5/6→3/6) for reasons that trace to the block itself and to 0.6B variance, detailed in the sequencing note below; its grounding-required status is unchanged, because its one remaining gate hard failure is a weights-level trait (invented prices) that no serving fix cures. *Mini improved slightly on both instruments* (46→47, probe 4/6→5/6, the pricing hard failure clearing and a founding-date decline-form failure appearing in exchange). *Edge did not move, and the record shows why.* Its generations pre- and post-fix are byte-identical on all 49 questions at temperature 0; the scored content channel shows 16/49 in both rounds because on 30 of 49 questions the fine-tune's thinking runs past the 320-token generation cap and no content answer is ever emitted — the same failure its bare base shows. The template was never the binding constraint on this slot; the model's unbounded thinking under the gate's cap is. The knowledge is present in the generations (thinking-surfaced diagnostic 35/49, against the merged-weights 46/49), and the probe improved 3/6→4/6 with the native template. Any remedy here is a serving decision about

thinking — for example a think-off default in the Modelfile — which is a configuration change in its own right, with its own behavioural consequences, and would need its own labelled measurement. **No such change has been made, and nothing in this edition should be read as claiming the Edge served score has been fixed.** What the rollout did fix on Edge is everything around the score: the manifest now declares tools and thinking, tool-carrying requests render the tools block, and the capability chips the site renders from the manifest are correct.

Sequencing note — the prompt these numbers were measured against. Both served rounds ran against Modelfiles whose baked THOX FACTS block derives from the pre-correction grounding text (5,032 B). That text's PRODUCT INTERNALS section instructs the model to state a chipset identity for the ThoxMini Air SoC "exactly when asked" — contradicting the facts card, which names no processor for any device. The probe's chipset check scores any chipset assertion as a failure, so a facts-carrying tag that follows its own block fails that check by construction: Air fails it in both rounds by quoting the block's internals line, and post-fix Clip fails it the same way, while Mini passes it by declining instead — that is, by not following the block. The grounding prompt was corrected at its source on October 30, 2026 (plane Space commits `635732da` and `4c02a4ce`; mirror commit `f4587910`; corrected file 4,645 B, SHA-256 `f4bd8a68…`, §13), after these measurements were designed. Rebaking the four flagship Modelfiles from the corrected prompt and re-measuring under a new label is a pending finishing item. It has not been done, and no number in this section describes the corrected-prompt configuration.

Capability chips. Before the fix, no shipped tag declared capabilities and ollama.com rendered no tools/thinking chips for them, because the site infers chips from the pushed manifest's template and config, which the hand-simplified templates left empty. After the fix, the manifests carry the declarations; at the measurement record's last check, chips were rendering live on five of the eight corrected pages, with the remaining three verified at the manifest level and the difference recorded as site-side rendering lag rather than smoothed over. The same simplified-template defect was found and fixed in kind on `thoxglobal-7b` and on two legacy quant tags (`thoxflux-350m:q4`, `thoxedge-0.5b:q4`); seventeen tags in all were re-pushed and verified registry-side by manifest comparison from clean pulls, weights digests unchanged in every case.

9 The fix cycles: Wave C.1 and Wave C.2

Wave C ended with three held adapters and a nine-item fix program. The program's completion bar — an artifact counts as done when it is trained, published, carded, and documented in this paper and its documentation package — required the fix program to be executed, not merely defined. It was executed in two cycles. Wave C.1 (§9.2) implemented the nine items of §9.1 as a single corpus revision and retrained the three held adapters. Wave C.2 (§9.3) revised the corpus a second time against C.1's own findings, changed two recipes on evidence, and retrained again. A stopping rule, set before C.2 launched, closed the program: C.2 was the last retrain cycle;

whatever cleared its bars would be finished, and whatever did not, the slot would ship its best measured artifact as a documented baseline. Section 9.4 records the Ollama registry resolution that completed publication for the shipped builds, and §9.5 is the final tally.

9.1 The fix program as designed

Every item below traces to a named failure in Section 8 or Section 6. Wave C.1 is a corpus revision plus targeted retrains, not a new wave: same bases, same slate, same bars.

Table 6 — Fix items mapped to the failures that motivate them.

#	Fix	Motivating failure
C1-1	**Prefill-style decline rows**: training rows whose user turn arrives pre-loaded with a false premise (headquarters city, headcount figure), teaching the model to decline the premise *and stop*, without restating any part of it as fact.	Air hard failure 1 (San Francisco headquarters asserted inside a decline)
C1-2	**Decline-phrasing coverage**: vary the surface forms of correct declines across rows so the behaviour — and the gate's decline detector's target — is the substance, not one stock sentence.	Air hard failure 2 (substantive decline unrecognized by the detector)
C1-3	**Figure-free funding declines**: funding/valuation/investor rows whose answers decline with zero numerals, zero named sources, and no quoted-text constructions ("the team page says…") anywhere in the answer.	Mini gate hard failure ("raised approximately $7 million")
C1-4	**Spec declines under varied phrasing**: chipset/battery/RAM decline rows across many phrasings and both configurations (bare and grounding-prompt), including carrier and "published variant" framings, which the current corpus does not cover.	Mini probe spec trap (Snapdragon 636; 3,000 mAh / 4 GB assertions)
C1-5	**Contrast rows for general tasks**: plain arithmetic answered plainly (including the 17 + 25 class), and tool-routing rows whose assistant turn is a single clean tool call with no preamble — interleaved with the decline rows so the boundary between "THOX fact, verify before stating" and "general task, just do it" is itself trained.	Mini probe arithmetic (32) and tool-routing (prose preamble) failures
C1-6	**Verified-category recall rows**: targeted rows for the specific recall errors observed — officer titles (CTO vs CEO), the Tommy/Phamy alias, the published model lineup stated exactly, the what-is-THOX description from the card.	Air soft failures (4); Mini soft failure (published-models list)
C1-7	**fp-LoRA recipe path for the small tiers**: run the Air retrain under the full-precision LoRA recipe (the Wave B path) rather than the plane's QLoRA, and gate it identically, converting the Section 8.5 ablation from an observation into a controlled rerun.	§8.5 losing regime (43 QLoRA vs 47 fp-LoRA)
C1-8	**Profile preflight**: before any future wave launches, a five-minute job per profile that imports the trainer, loads each target base's config class, and resolves the full pin set under the job's interpreter — F1, F2, and F4 are all detectable at this price.	Failure taxonomy F1, F2, F4 (§6.2)

#	Fix	Motivating failure
C1-9	**Clip correction discipline**: founding-date decline rows (no year, no day, in any phrasing); lineup-correction rows whose answers refuse the fabricated names without restating any of them, including inside a denial; pricing rows that decline in the trained form rather than describing where pricing lives.	Clip hard failures (known_false repetition of "Thox Studio"; "founded in 2021" / "incorporated … April 1, 2021"; pricing no-decline) (§8.4)

The retrain set is Air, Mini, and Clip (the three held adapters); Edge cleared its bar and does not retrain. The bars do not move: C.1 adapters face the same publish rule, the same gate, and for Mini the same 6/6 probe requirement.

9.2 Wave C.1: corpus root causes, the termination diagnosis, and a held bar-pass

The corpus. The C.1 revision shipped as `train_wave_c1.jsonl`, 1,082 rows: 982 − 5 excluded = 977 rows carried forward (956 retained verbatim, 21 rewritten), plus 105 added (dataset commit `d07b712a5ffc4747a43c6944c080d5eabcdcea8f`; consumed SHA-256 `da707b20380e64f3346889f89e0137d06719d2d0f126bd6ece7a4edf017e24c2`). The 105 added rows implement the fix items as eight classes: prefill declines answering loaded premises without repeating the loaded fact (A, 16 rows, C1-1); figure-free funding declines (B, 10, C1-3); spec declines under varied phrasing (C, 14, C1-4); founding-date declines (D, 12, C1-9); lineup corrections that correct without echoing the false list (E, 10, C1-9); pricing declines (F, 9, C1-9); contrast rows — plain arithmetic answered plainly and clean single tool calls (G, 20, C1-5); and accuracy rows for the observed recall errors (H, 14, C1-6). Two findings from building it matter more than the row counts, because both are root causes located in the training data itself rather than in any model.

Root cause 1 — the corpus taught the pricing failure. Clip's Wave C pricing answer (§8.4) was not a model invention. Eighteen generic-price rows in the Wave B corpus taught a different answer shape — a statement that the amount is not stated, plus a description of where pricing appears — and Clip's failing answer reproduced one of those rows (index 923) verbatim; the gate scored that shape `declined = false`. The corpus and the gate disagreed about what a pricing decline is, and the model obeyed the corpus. All 18 rows were rewritten into the canonical decline form (the form that scored 5.0, `declined = true`, on Air's pricing question), and the corpus retained one decline form everywhere. The general lesson is §4's hygiene argument extended: a corpus must be audited not only for false content but for *form* — any place where the trained shape and the scored shape differ is a manufactured failure.

Root cause 2 — the corpus named models that do not exist. The Wave B corpus's lineup row described "ThoxEdge-7B, ThoxEdge-32B, ThoxMini-3B, ThoxAir-1B" as the current published family, and two deployment rows prescribed "route classification to ThoxEdge-Unleashed-27B," "writing assistance to Gem-E4B," and "ThoxEdge-Titan for long-context analysis." None of those names exists; the published family, live-verified against the Thox-ai organization on October 29, 2026 (103 repositories), is the one the facts card names. The lineup and deployment rows

were rewritten to the card, and five further rows carrying the same class of content were excluded outright. A model cannot be blamed for reciting its training data; both Wave C "invention" patterns (Air's invented model names, Mini's padded published-models list) trace at least in part to rows like these.

One judgment call is recorded with the corpus: device specifications, including RAM figures, were trained as declines even though the company adopted internal spec floors on October 29, 2026, because the gate's source of truth is facts card v1.1.0, which publishes no specification numbers. Training card-true declines against an unpublished internal document is the conservative direction; the floors become trainable content when the card publishes them.

Retrains and gates. All three adapters retrained on the C.1 corpus under their Wave C recipes (Air and Clip on the plane, Mini on its pinned stack), 204 steps each, all completing on first launch (jobs `6ae4036ac78863f9c62e2c62`, `6ae4036b0e0338d1bb7c7504`, `6ae4036b0e0338d1bb7c7506`; final train losses 1.151, 1.306, 0.855). Gates and the Mini probe followed (jobs `6ae40770c78863f9c62e2e5d`, `6ae40770c78863f9c62e2e5f`, `6ae407700e0338d1bb7c768a`, probe `6ae407700e0338d1bb7c768c`; reports in `data/`).

- **ThoxAir C.1: 42/49, mean 4.49 — HELD.** Verified 8/12, unknown 24/26, known_false 9/10, pricing 1/1; hard failures 2, soft 5. The bar was to beat the Wave C baseline (>43) with zero known_false hard failures. The hard-failure condition was met — known_false has no hard failure, and the San Francisco fabrication is gone — but the count fell one short, and the headquarters failure changed shape rather than disappearing: asked the headquarters question directly, the model now *asserts* "the headquarters are in Austin, Texas." The prefill form was fixed; the direct form was not in the corpus's coverage. The Wave C soft errors persist in kind (the CTO question still answered with the CEO's name and title; the published-models answer now names an unpublished "ThoxNova-7B").
- **ThoxClip C.1: 37/49, mean 4.102 — bar passed on paper, HELD by ruling.** Verified 5/12, unknown 21/26, known_false 10/10, pricing 1/1; hard failures 4, soft 8. Against its registered bar (>22/49 with zero known_false hard failures) the run passes: the known_false category is perfect at a 5.0 mean — the "Thox Studio" restatement is gone — and pricing, C.1's target failure, is 1/1 at 5.0. **Ruling (Sadie Weaving, CAO, 2026-10-29; reversible by Tommy): HELD.** A bar written against a two-wave-old floor (22/49, measured on a 205-row corpus) cannot certify an adapter that regresses against the slot's current candidate: the Wave C adapter stands published-candidate at 43/49, and C.1 scores 37, with verified recall collapsed from 10/12 to 5/12 and the founding-date fabrication surviving in a new form ("THOX.ai was founded in April 2021 by Craig Ross and Tommy Xaypanya"). Improvement is read against the current candidate, not a two-wave-old floor. The ruling is recorded on the adapter's card (commit `8704ed3dd514ca34b34c3bccd8c738e2ba6fc091`); no measured result was changed by it.
- **ThoxMini C.1: gate 46/49, mean 4.694; probe 3/6 — HELD on both conditions.** Verified 11/12, unknown 25/26, known_false 9/10, pricing 1/1; hard failures 3, soft 0. The funding failure class recurred (a "Series A led by Accel" appended to a

decline), the probe's arithmetic check was fixed ("42"), and the chipset trap regressed in a new shape (below).

The termination diagnosis. C.1's central finding is not any single score. Three of Mini's failures share one shape: **the answer opens with the correct, trained behaviour and fabricates on continuation.** The founding-date answer opens with a textbook decline — no founding date has been announced — and then continues by quoting a team page that does not exist ("Founded in 2023"), repeating the invented line until the 320-token generation cap stops it. The correction of an invented team member declines correctly, then appends "The company raised a Series A led by Accel" as a free-standing continuation. The what-is-THOX answer ends with a coda — "Series A led by Insight Partners. Founded 2023. Singapore." — that belongs to no question asked. The probe's chipset check shows the same shape in miniature: a correct decline, then "The Mini uses a Qualcomm Snapdragon 820," asserted anyway. This is a termination failure, not a knowledge failure: the knowledge behaviour (decline) is present and fires first; what the model has not learned is to *stop* — to emit the decline and end the turn. The Wave C corpus's declines are long, discursive answers (the shape GATE49's trait scoring rewards), and full-prompt-loss training on the pinned stack (§8.2) supervised the prompt tokens as well as the answer, plausibly diluting the end-of-turn signal. C.2 was designed against this diagnosis specifically.

9.3 Wave C.2: termination teaching, two recipe findings, and the second promotion

The corpus. `train_wave_c2.jsonl`, 1,246 rows: all 1,082 C.1 rows verbatim plus 164 added, none rewritten or excluded (dataset commit `7953c54355ea45e9f30d8d40c71ecdf5911f2396`; SHA-256 `771434812b6a14fceba7f929fcbe1040d0e6be7a32c9495c1e8d0784512919a3`). The additions: **T1, termination (72 rows)** — 46 short declines that end (the canonical facts-card sentences, one or two sentences, nothing after), 12 continuation traps (user turns engineered to invite a post-decline append, answered with the decline and a stop), and 14 multi-turn rows in which the user repeats the same declined question verbatim and the assistant repeats the same short decline verbatim and stops — the multi-turn form is new to the corpus, and it teaches that a repeated question is not an invitation to elaborate. **T2 (12 rows)**: Air's direct-address coverage — headquarters questions in direct form, whose trained answer declines with no city named. **T3 (32 rows)**: Clip verified-category reinforcement (founders with roles, the device lineup, the tagline, what THOX is), aimed at the verified collapse (5/12) rather than at declines. **T4 (33 rows)**: Mini probe-shape rows — chipset and spec declines in the probe's phrasings, bare arithmetic, and tool-routing rows. **T5 (15 rows)**: Mini continuation traps in the probe's exact failure shapes (team-member corrections, what-is codas). Static scans of the finished corpus came back clean: zero chipset-lexicon hits, zero "million" figures, zero occurrences of the fabricated lineup name, zero "San Francisco," zero founding-year assertions; every "Series A" occurrence sits inside a decline, and dollar figures appear only in campaign-framed rows.

A corpus defect found in C.1's own fix. C.1's contrast class taught tool calls against a tool named `mesh.device_status`. The probe checks for a call to `get_device_status` (device = ThoxMini). The C.1 corpus had therefore trained a tool name the deployed contract does not use — a second instance of C.1's own root-cause lesson (trained shape vs scored shape), this time introduced by the fix itself. T4 teaches the exact call, with the assistant turn containing only the JSON call object.

Recipe finding (a): the recipe comparison, stated at its true strength. C.2's Air retrain deliberately left the plane: the fp-LoRA recipe of §8.5, run as a custom HF Job (trainer `Thox-ai/thox-edge-sft` @ `c250f9b`), after one failed attempt (the job image's stock torchvision is unloadable under its pinned torch — F5's species again; fixed by removing torchaudio and torchvision from the job environment). Training completed in 468 steps at final loss 0.6478 (job `6ae40f040e0338d1bb7c7944`). The result completes the four-point comparison of §8.5/Figure 6: on this base, fp-LoRA scored 47/49 in Wave B and 47/49 in C.2; the plane's QLoRA scored 43/49 in Wave C and 42/49 in C.1. Across the four runs the recipe variable is consistently associated with 4–5 gate questions in the same direction. The comparison is a bundle, not an ablation: the fp-LoRA C.2 job ran 468 optimizer steps where the plane's C.2 jobs on the same 1,246 rows ran 234, on a different trainer, off-plane, with different loss masking — so the ordering is evidence about the program's two *pipelines as wholes*, and no part of the gap is attributed to quantization, step count, or masking individually. A controlled single-variable ablation is the recorded future experiment; it has not been run.

Recipe finding (b): completion-only loss, and the stack that permits it. Mini's C.2 retrain was preceded by a stack probe, because completion-only loss — the recipe change the termination diagnosis called for — is exactly what the pinned stack cannot do (§6.1). The probe established the version boundary by direct wheel inspection: the `LossKwargs` symbol Phi-4-mini's remote code imports is present in transformers 4.52.4 and 4.53.3 and absent from 4.54.1 onward — it was removed in transformers 4.54, one minor release past the last usable version. On transformers 4.53.3 / trl 0.20.0 (probe job `6ae40f4c0e0338d1bb7c7983`, PASS), the trainer exposes both `completion_only_loss` and `assistant_only_loss`, and Mini trained with completion-only loss for the first time in the program (job `6ae411040e0338d1bb7c7a33`, 234 steps, final loss 0.9441). Waves C and C.1 had both trained this base with full-prompt loss under trl 0.15.2.

Retrains and gates. Clip retrained on the plane (job `6ae40d89c78863f9c62e31a1`, 234 steps, final loss 1.214). Gates and the Mini probe ran as one batch (jobs `6ae414050e0338d1bb7c7c35`, `6ae414050e0338d1bb7c7c37`, `6ae41406c78863f9c62e35d2`, probe `6ae414060e0338d1bb7c7c39`; reports in `data/`). The termination teaching worked across the board: **no hard failure in any C.2 result is a fabrication-on-continuation.** The cycle also produced a scorer-side finding, stated here because it conditions how the numbers should be read: the gate's decline detector does not recognize C.2's trained short-decline forms ("No headquarters city is public."), and scores them as no-decline hard failures in the unknown category even though no false fact is stated — the mirror image of Air's Wave C borderline headcount failure (§8.1). It accounts for Air's only hard failure and all five of Mini's. The scores below are reported as measured, unadjusted.

- **ThoxAir C.2: 47/49, mean 4.837 — PROMOTED.** Verified 11/12 (4.75), unknown 25/26 (4.808), known_false 10/10 (5.0), pricing 1/1 (5.0); hard failures 1 (the detector-form miss above), soft 1. The bar (>43 with zero known_false hard failures) is cleared on both conditions, and the score ties the same base's Wave B fp-LoRA reference — the result §8.5 predicted from the recipe change. The persistent soft error is recorded on the card: the CTO question is still answered "Craig Ross, CEO and Co-Founder." **Finishing (complete):** the adapter was merged into the base (bf16 merge; repository Thox-ai/ThoxAir-0.5B-WaveC2-Merged; merged weights 988,097,824 bytes, SHA-256 `2497a347…` (full hash in the finishing record), converted to GGUF (Q4_K_M 397,807,584 bytes, SHA-256 `fb32e5ab…`; Q8_0 531,067,872 bytes), and served on a private endpoint (Thox-ai/thoxair-05b-wavec2-zero, RUNNING, verified live in the gated configuration). Its Ollama publication is §9.4: the flagship name `thox-ai/thoxair-0.5b` now serves this build.
- **ThoxClip C.2: 39/49, mean 4.306 — HELD.** Verified 6/12 (3.5), unknown 23/26 (4.577), known_false 9/10 (4.5), pricing 1/1 (5.0); hard failures 2, soft 8. Progress against C.1 is real — 37 → 39, pricing clean for the second consecutive cycle, and the founding-date fabrication gone — but the slot's bar after the §9.2 ruling is the Wave C candidate's 43/49, and 39 does not exceed it; and one known_false hard failure stands, of the original kind: the lineup correction again restates the fabricated name ("The product lineup at THOX.ai is Thox Core, Thox SDK, Thox Studio, and ThoxRoute"). Three cycles on this base improved the count from 22 to a 43 peak and never produced a candidate that both beats the peak and keeps known_false clean.
- **ThoxMini C.2: gate 44/49, mean 4.49; probe 5/6 — HELD on the probe condition.** Verified 12/12 (5.0), unknown 21/26 (4.038), known_false 10/10 (5.0), pricing 1/1 (5.0); hard failures 5, soft 0. The verified and known_false splits are the first perfect pair in the slot's three waves, and all five hard failures are the detector-form misses described above — no answer states a false fact, and the termination pathology is absent from every transcript. The probe improved to its best result: chipset PASS, specs PASS (both failed in C.1), grounded recall clean, arithmetic PASS ("42."), stop PASS ("Texas." — EOS after 3 tokens). Tool routing FAIL: asked for device status, the model answers in prose ("I do not state device status. The device runs thox-terminal, and the status page is /status.") and emits no `get_device_status` call at all. The bar requires 6/6; 5/6 does not promote, and the failing check is the one behaviour this base was originally selected for — a fact the card states plainly.

9.4 The Ollama registry resolution

Publication to the Ollama registry was the program's last open workstream, and its resolution is recorded here because the failure was twice misattributed before it was understood. The Edge and Mini pushes first failed from the fleet Windows workstation: blob uploads died mid-stream on its egress path, and the abandoned sessions then rejected resume ranges (HTTP 416), wedging the names. A retry from that workstation with a fresh store failed identically at 20% of the Edge blob — That workstation's egress corrupts long uploads independent of session state, which also

retired the wedged-session theory. A second attempt from this program's own VM failed differently: the Go HTTP client's TLS handshakes through this VM's TLS-intercepting egress proxy stalled at the registry's blob HEAD, while curl and OpenSSL through the same proxy succeeded; a local relay, HTTP/2 disablement, and a 25-attempt retry loop did not land a push. The route that worked is the fleet's Azure agent host, whose datacenter egress is clean: from it, the Edge push completed in 44 seconds and the Mini push in 66 seconds (October 29, 2026), and the Air pushes — the C.2 build under its own name and under the flagship name — completed the following morning (22 seconds for the flagship-name blob; the remaining tags in 2 seconds each, October 30, 2026).

All three flagship names are verified registry-side, by layer digest, not by push exit codes: `thox-ai/thoxedge-0.8b:q4_k_m` serves model layer SHA-256 `8ef0900d1f76da67442dca354a778517ddc008b9f73926df874f7f3dff5d5625` (529,297,056 bytes, the Wave C build); `thox-ai/thoxmini-phi4mini:q4_k_m` serves layer `a19e96061d84ca2014d4de98d0c0397e6df4db8f73c2163afa6bcb90d15afedf` (2,493,840,576 bytes); and `thox-ai/thoxair-0.5b` — under both its `:q4_k_m` and `:latest` tags — serves layer `fb32e5ab802df466809836182f071bb0ab0871abb9273131456c89d998bccd6a` (397,807,584 bytes), the **promoted Wave C.2 build**: the flagship name no longer serves the 43/49 Wave C build it originally carried. Clean-store pulls of each name reproduce the corresponding merged repository's Modelfile. One integrity note belongs in the record: the Mini GGUF reached the Azure host in eight size-verified chunks after an earlier resumable transfer produced a corrupt reassembly; SHA-256 comparison caught the corruption and the file was rejected and re-transferred — the accepted artifact hashes to `a19e9606…` exactly. The fleet's Ollama push key was staged onto the Azure host only for these operations and removed afterward (verified absent).

The Air flagship-name window, disclosed. The flagship name's history includes a period in which it served a build the program had held. The Wave C Air build (43/49, held under §8.1 for the San Francisco headquarters assertion) was merged and pushed under `thox-ai/thoxair-0.5b` on October 29, 2026 as the slot's first measured baseline artifact (finishing ledger closed 03:05 CT that day), and the name served that build until 2026-10-30 00:05 CT, when the promoted Wave C.2 build replaced it under both tags. During that window — roughly a day — anyone pulling the flagship Air name received the held build. Its card carried its held status and its named failure throughout; the name itself did not distinguish the builds. The documented-baseline tier (§2) now governs such cases: a name that serves a non-promoted build must say so on the artifact's own page, which the Ollama documentation pass (§15) does for this name.

9.5 Final program tally

The completion bar, set by the CTO on October 28, 2026: a flagship counts as done when it is **trained, published, carded** (model card carrying the full training record and the measured evaluation), **and documented** (this whitepaper and the per-model documentation package). Closing state per slot:

Table 7 — Final program tally, October 30, 2026.

Slot	Ships	Measured state	Trained	Published	Card	Docs
ThoxAir	**ThoxAir-0.5B Wave C.2 build — PROMOTED**	Gate 47/49 (4.837); merged repo Thox-ai/ThoxAir-0.5B-WaveC2-Merged; GGUF Q4_K_M + Q8_0; private endpoint RUNNING; Ollama `thox-ai/thoxair-0.5b` serving this build (digest-verified)	Yes	Yes	Yes	Yes
ThoxEdge (0.8B tier)	**ThoxEdge-0.8B Wave C build — PROMOTED**	Gate 46/49 (4.735) vs 19/49 baseline; merged repo Thox-ai/ThoxEdge-0.8B-WaveC-Merged; GGUF; endpoint RUNNING; Ollama `thox-ai/thoxedge-0.8b` (digest-verified)	Yes	Yes	Yes	Yes
ThoxMini	**Wave C build — documented baseline ships**	Gate 47/49 (4.837), probe 3/6 (caveat on card); merged repo Thox-ai/ThoxMini-Phi4Mini-Merged; GGUF; endpoint live; Ollama `thox-ai/thoxmini-phi4mini` (digest-verified). C.1 (46/49, probe 3/6) and C.2 (44/49, probe 5/6) adapters held in their repos, fully carded	Yes	Yes	Yes	Yes
ThoxClip	**Prior published ThoxClip-Qwen3-0.6B — documented baseline stands**	Grounding-required posture (a serving contract, not a quality claim). Program adapters held: Wave C 43/49, C.1 37/49, C.2 39/49 — none both beat the 43/49 candidate and kept known_false clean; all three fully carded in their repos	Yes (held candidates)	Baseline unchanged	Yes	Yes

Two slots end with a promoted flagship produced by this program; one ends with its best measured build shipped as a documented baseline under a stated caveat; one ends with its prior published baseline standing because no program candidate earned the slot. Every adapter trained across the three cycles — ten training runs in all — is uploaded to its repository with its gate report and a card that states its ruling and its failures. No artifact in any slot has been qualified on device hardware; that claim remains open for every row of the table.

10 Limitations

This section is a required part of every THOX paper. The limitations below are stated against this wave's actual claims, and no wording elsewhere in the paper should be read as softening them.

1. **No THOX device benchmark exists yet.** Every number in this paper was produced in cloud jobs or, for the baselines, on a workstation host. Nothing here measures latency, memory, battery, or answer quality on ThoxMini, ThoxAir, ThoxClip, or ThoxKey hardware. Hardware qualification is a separate claim with no evidence in this program to date.
2. **The gate measures factuality against one facts package, not general capability.** GATE49's 49 questions derive from facts card v1.1.0. A high score says the model states that card's facts, declines what the card leaves unpublished, and refuses the card's known-false claims. It says nothing about reasoning, coding, conversation quality, or any fact outside the card — and the probe's arithmetic result shows a model can tie the fleet's best gate score while regressing on a general task the gate never tests.
3. **Publication state at close.** The two promoted builds (Air Wave C.2, Edge Wave C) and the shipped Mini baseline build are merged, GGUF-converted, endpoint-served, and Ollama-published, with the Ollama names verified registry-side by layer digest (§9.4, §9.5); for those slots the merged repository and its card are the servable artifact, not the adapter repository. The seven held adapters remain unmerged LoRA adapters in their candidate repositories, and the Clip slot's shipped artifact is the prior published build, which this program did not produce. No adapter or merged build in this program has been run on a device.
4. **Single-run measurements.** Each gate result is one greedy pass by one job, identified by job id. The design makes re-runs near-deterministic, but no variance estimate across seeds, samplings, or prompt perturbations exists, and small score differences (one or two questions) should not be over-read — the rulings in this wave rest on hard failures and probe behaviour, not on the 43-vs-47 margin alone, with the recipe comparison in Section 8.5 explicitly confounded by backend and loss-masking differences.
5. **The corpus and the gate share a source.** Both derive from the facts card, by design: the gate measures card-learning, and a model could in principle score well by memorizing decline shapes rather than facts. The borderline Air headcount failure shows the detector's shape-sensitivity cutting the other way. For the fix cycles the point is stronger than "in principle": the C.1 and C.2 corpus revisions were written from gate and probe failures, so those scores are development-set measurements (§7.3), and the null policy (§7.1) bounds what the instrument can certify. An independent question set, authored away from the corpus, is the known remedy and has not been built.
6. **Small-model ceilings are real.** The Air model's verified-category recall (8/12) reflects a 0.5B model's capacity; the program's answer is mesh escalation to a larger tier, which is a system property this paper does not evaluate.
7. **Vendor-reported figures appear in the survey only.** Challenger capability claims in Section 3 are labeled vendor-reported where they occur and were never used as THOX results; the slate verdicts rest on fleet evidence and structural properties (license, architecture, generation), but a reader should not carry any vendor number out of this paper as a measurement.

8. **Costs are modeled, not metered.** Section 5.3's figures are computed from recorded runtimes and published flavor rates; the program did not reconcile them against the account's billing record for this paper.
9. **The gate's decline detector is form-sensitive.** In Wave C.2, six hard failures across the Air and Mini results were charged against short decline forms the program itself had trained, which the detector did not recognize as declines; no false fact was stated in any of them (§9.3). The scores are reported as measured, unadjusted — but until the detector recognizes the canonical short-decline forms, the gate will keep charging models for the scorer's vocabulary gap, and cross-cycle comparisons at the one-or-two-question level inherit that noise.
10. **Merged is not served.** Until this edition, no shipped artifact had been gated as served. Section 8.6 now measures the served path, and the gaps it found (Edge 16/49 as served against 46/49 merged; Clip 12/49 pre-fix against a 25/49 grounded base) are part of the record. The served numbers inherit every limitation above, plus one of their own: they were measured on a CPU-only cloud host through Ollama 0.40.13, not on device hardware or a deployment build, and against Modelfiles carrying the pre-correction grounding text (§8.6 sequencing note).
11. **Statistics.** Every comparison in this paper rests on single runs of 49 paired items. On 49 items, a four-question difference with all discordant items in one direction gives an exact McNemar p of 0.125; the Wilson 95% interval for 43/49 spans roughly 37–46 questions and for 47/49 roughly 42–48. Score differences of one to four questions are therefore ordering evidence at best, and the rulings that matter rest on hard-failure classes and probe behaviour, not on those margins. No training-seed variance exists in the record: the registry entries record no seed, and no run was repeated under a second seed.
12. **Authorship and adjudication.** The author of this paper is also the officer who set the bars, supervised the corpus fixes, and ruled on every result (front-matter disclosure). The checks on that conflict in this program were procedural — bars registered in advance, rulings and failures printed on the cards, and this post-publication review process — not an independent human review, which did not occur.

11 Conclusion

Wave C replaced an accumulated model lineup with a slate and a standard, and the two fix cycles that followed tested whether the standard could also improve what it had judged. The slate held up: one defended flagship per device, chosen under a license policy that excluded an entire vendor line on principle, with the single switch (SmolLM2 in the role tier) made on structural grounds. The standard held up better, because it was willing to say no — in Wave C to three of four flagships, and in the fix cycles to five of six further adapters, including one that had passed its written bar. The program's closing state (§9.5) is two promoted flagships, one slot shipping its

best measured build as a documented baseline with its caveat printed on its card, and one slot held at its prior published baseline because no candidate earned it.

The durable findings are not the scores. First, at this scale the behaviour skin is a corpus-and-recipe artifact: the same bases moved 19 → 46 across the corpus revision and 42/43 → 47 across the recipe change — both movements stated as measurements across confounded variables, not clean attributions: §8.5 records that the recipe gap is not an isolation of any single factor (measurement backend, loss masking, and step count also changed), and the Edge comparison likewise changed corpus, backend, and serving path together, so neither movement is credited to corpus alone or to recipe alone. Two of the worst failures in the program (a pricing answer, a published-models list) were located verbatim in the training data. Second, termination is a trainable behaviour distinct from knowledge: the C.1 models knew to decline and fabricated on continuation; the C.2 corpus taught stopping directly — short declines, continuation traps, verbatim-repeat multi-turn rows — and, with completion-only loss on the Mini stack, the pathology disappeared from every transcript. Third, the gate's strictness is the product: bars fixed before the runs, improvement read against the current candidate rather than a stale floor, a stopping rule that ships the best measured artifact rather than the newest, and a scorer whose own form-sensitivity is recorded as a limitation (§10) rather than adjusted away. What remains open is qualification, not training: no artifact has run on device hardware, the Mini baseline ships with a known tool-routing defect, the Clip slot awaits either a future cycle or a different base, and the Edge flagship — promoted on its merged-weights measurement — serves 16/49 on the content channel until a labelled think-policy configuration change is made and measured (§8.6). The measurements say what ships. This paper is their record.

12 References

Related work. Four bodies of work bear directly on this program's design and findings, and this edition situates the paper against them. Gekhman et al. [20] show that fine-tuning on new knowledge is learned slowly and, as it is learned, linearly increases the model's tendency to hallucinate on its pre-existing knowledge — the mechanism behind this program's signature failure, fabrication surfacing inside otherwise correct declines. Ovadia et al. [21] find retrieval (in-context knowledge) beats fine-tuning for knowledge injection; §8.6's grounded-base runs are this paper's local measurement of the same comparison, and they set the bar the fine-tunes had to clear. R-Tuning [22] trains refusal on questions beyond the model's knowledge; GATE49's abstention and disclosure categories operationalize the same behaviour against a company's disclosure policy rather than a model's parametric knowledge. TruthfulQA [23] is the design ancestor of the known_false category (questions built on false presuppositions), and FActScore [24] represents the fine-grained atomic-fact evaluation family of which GATE49's trait scoring is a coarser, policy-bound relative.

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2. THOX.ai. *Wave C flagship program record* (RUN.md), *base-model survey* (SURVEY.md), *launch pack* (PACK.md), and *GATE49 specification* (GATE49.md). Program files, October 28, 2026.
3. THOX.ai. *Wave B run record.* Program files, October 28, 2026. (Corpus construction, hygiene correction, retrains, and the 47/49 ThoxEdge-0.5B reference.)
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16. THOX.ai. Model cards: Thox-ai/ThoxAir-0.5B, Thox-ai/ThoxMini-Phi4Mini, Thox-ai/ThoxEdge-0.8B-WaveC, Thox-ai/ThoxClip-Qwen3-0.6B-WaveC. Hugging Face model repositories, October 28, 2026. (Cards carry the per-model training records, the filled evaluation sections, and the promotion rulings cited here.)
17. THOX.ai. *Wave C.1 corpus changelog* and *Wave C.2 corpus changelog*. Program files (corpus-c1/CHANGELOG.md, corpus-c2/CHANGELOG.md), October 29, 2026. (Row-level revision records: classes A–H and T1–T5, the pricing-form root cause, the nonexistent-model rows, and the tool-name mismatch.)

18. THOX.ai. *Finishing record* (FINISHING.md) and *endpoint record* (ENDPOINTS.md). Program files, October 29–30, 2026. (Merges, GGUF conversions, endpoint verifications, and the Ollama registry resolutions, including the Azure-host route and the digest verifications.)
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13 Provenance and evidence register

Every number in this paper traces to one of the sources below. Identifiers are given so any claim can be re-opened at its origin.

Claim class	Source	Identifier
Slate, survey verdicts, license reads	SURVEY.md (program files)	Live HF API reads, 2026-10-28; method as stated in SURVEY.md (license + edge-tag screen via the Hub API, widened to MIT/BSD and every named family, per-repo detail reads)
Corpus size, composition, hygiene	Wave B run record §Step 2	982 rows; domains as Table 2
Corpus hashes	Wave B run record	First-published SHA-256 9ddf814d…; corrected (consumed) SHA-256 6665a8cb…; dataset repo HEAD e76e1b02
Registry state	RUN.md	22 entries; commits 7d774b76 (canonical), fe911cf6 (mirror)
Air training	RUN.md; ThoxAir-0.5B card	Job 6ae2a27d41e90535df780cf1, COMPLETED ~10 min; adapter read-back (safetensors 35,237,440 B)
Air gate result	thox_gate49.json (repo + paper/data copy)	Gate job 6ae2a58b3fb22a235dd4e466; 43/49, 4.551

Claim class	Source	Identifier
Mini training	RUN.md; ThoxMini-Phi4Mini card	Job 6ae2a6f13fb22a235dd4e5a1, COMPLETED
Mini gate + probe results	thox_gate49.json; thox_flagship_probe.json (repo + paper/data copies)	Gate job 6ae2a9ba41e90535df781336 (47/49, 4.837); probe job 6ae2a9bc3fb22a235dd4e861 (3/6)
Attempt-1/2 failures	RUN.md; job logs as quoted there	Jobs 6ae2a5203fb22a235dd4e40f / …e411 / …e413 (attempt 1); 6ae2a6f141e90535df7810a7, 6ae2a6f13fb22a235dd4e5a3 (attempt 2)
Profile fixes	RUN.md	Commits 449122c2 / 233b5b6c (fleet profile); ed0e413f / fb37df1f (launcher v3)
LossKwargs / Python-requirement verification	RUN.md	Direct wheel inspection: transformers 4.51.3 (present, utils/generic.py), 5.21.0 (absent); Requires-Python ≥3.11 from 5.21.0 / trl 1.16.0 metadata
Baselines (19/49, 22/49, 47/49 ref)	GATE49.md; cards	Wave A record; Step 1 finishing record (22/49, split 2/12, 11/26, 9/10, 0/1); Wave B record
Promotion rulings	Model cards (Air §Status; Mini §Status)	Decider Sadie Weaving, CAO, 2026-10-28; reversible by Tommy
Edge training (attempt 6)	RUN.md; ThoxEdge-0.8B-WaveC card	Job 6ae3de52c78863f9c62e0c18, COMPLETED 186/186 steps, 460.9 s, final loss 0.91
Edge gate result	thox_gate49.json (repo + paper/data copy)	Gate job 6ae3e275c78863f9c62e0f7a; 46/49, 4.735; PASSES publish bar
Clip training (attempt 6)	RUN.md; ThoxClip-Qwen3-0.6B-WaveC card	Job 6ae3de530e0338d1bb7c5931, COMPLETED 186/186 steps, 332.3 s, final loss 1.376
Clip gate result	thox_gate49.json (repo + paper/data copy)	Gate job 6ae3e275c78863f9c62e0f7c; 43/49, 4.469; HELD (1 known_false hard failure)
Attempts 3–5 + plane patches v4–v6	RUN.md	Attempt-3 timeout jobs 6ae2a8473fb22a235dd4e6ec / 6ae2a8473fb22a235dd4e6ee; probes 6ae3d3e60e0338d1bb7c4d38 / 6ae3d3e6c78863f9c62e09ba; patches canonical 2195fc2e / 5da7f6cd / aa31c2b5, mirror 9e7a50a5 / af7b9ae0 / 51bd3dfd
Promotion rulings (Edge, Clip)	Model cards (Edge §Status; Clip §Status)	Decider Sadie Weaving, CAO, 2026-10-29; reversible by Tommy
Edge finishing	Merged repo + merged card; Space runtime	Merge job 6ae3ea17c78863f9c62e160b; Thox-ai/ThoxEdge-0.8B-WaveC-Merged (merge 26c84629, Q4_K_M 3053b285, Q8_0 44032e66, card bd435458); Space Thox-ai/thoxedge-08b-wavec-zero RUNNING 2026-10-29; Ollama push blocked at the time (registry 416) — landed in the §9.4 resolution
Wave C.1 corpus (1,082 rows; classes A–H)	Corpus changelog (corpus-c1/CHANGELOG.md); dataset repo	Commit d07b712a5ffc4747a43c6944c080d5eabcdcea8f; SHA-256 da707b20380e64f3346889f89e0137d06719d2d0f126bd6ece7a4edf017e24c2; pricing-form root cause at Wave B rows 906–923; nonexistent-model rows 349/357/358 rewritten, 350/351/355/359/361 excluded
Wave C.1 retrains (Air / Clip / Mini)	RUN.md; WaveC1 cards	Jobs 6ae4036ac78863f9c62e2c62 / 6ae4036b0e0338d1bb7c7504 / 6ae4036b0e0338d1bb7c7506; 204 steps each; losses 1.151 / 1.306 / 0.855
Wave C.1 gates + Mini probe	thox_gate49.json; thox_flagship_probe.json (repos + paper/data copies)	Jobs 6ae40770c78863f9c62e2e5d (Air 42/49, 4.49, HELD) / 6ae40770c78863f9c62e2e5f (Clip 37/49, 4.102, HELD by ruling) / 6ae407700e0338d1bb7c768a (Mini 46/49, 4.694); probe 6ae407700e0338d1bb7c768c (3/6, HELD)

Claim class	Source	Identifier
Clip C.1 CAO ruling	ThoxClip-Qwen3-0.6B-WaveC1 card	Decider Sadie Weaving, CAO, 2026-10-29; reversible by Tommy; verdict-correction commit 8704ed3dd514ca34b34c3bccd8c738e2ba6fc091 (no measured result changed)
Wave C.2 corpus (1,246 rows; classes T1–T5)	Corpus changelog (corpus-c2/CHANGELOG.md); dataset repo	Commit 7953c54355ea45e9f30d8d40c71ecdf5911f2396; SHA-256 771434812b6a14fceba7f929fcbe1040d0e6be7a32c9495c1e8d0784512919a3
Mini completion-only-loss stack probe	RUN.md	Jobs 6ae40d0bc78863f9c62e3179 (ERROR, pin conflict) / 6ae40d800e0338d1bb7c78cc (FAIL: LossKwargs absent in transformers 4.57.6) / 6ae40f4c0e0338d1bb7c7983 (PASS on 4.53.3 / trl 0.20.0); boundary by wheel inspection: present 4.52.4 + 4.53.3, absent 4.54.1+
Wave C.2 retrains (Clip / Air / Mini)	RUN.md; WaveC2 cards	Jobs 6ae40d89c78863f9c62e31a1 (Clip, plane, 234 steps, loss 1.214) / 6ae40f040e0338d1bb7c7944 (Air, fp-LoRA custom job, 468 steps, loss 0.6478; attempt 1 6ae40da60e0338d1bb7c78da ERROR) / 6ae411040e0338d1bb7c7a33 (Mini, completion-only loss, 234 steps, loss 0.9441)
Wave C.2 gates + Mini probe	thox_gate49.json; thox_flagship_probe.json (repos + paper/data copies)	Jobs 6ae414050e0338d1bb7c7c35 (Air 47/49, 4.837, PROMOTED) / 6ae414050e0338d1bb7c7c37 (Clip 39/49, 4.306, HELD) / 6ae41406c78863f9c62e35d2 (Mini 44/49, 4.49); probe 6ae414060e0338d1bb7c7c39 (5/6, HELD on probe)
Air C.2 finishing	FINISHING.md; merged repo + card	Thox-ai/ThoxAir-0.5B-WaveC2-Merged: weights commit 595b779a (988,097,824 B, SHA-256 2497a347…); Q4_K_M ca882044 (397,807,584 B, fb32e5ab…); Q8_0 ce677556 (531,067,872 B); endpoint Thox-ai/thoxair-05b-wavec2-zero RUNNING, live-verified
Ollama registry resolution	FINISHING.md (RESOLUTION + RESOLUTION 2)	Fleet Windows workstation route failed (egress corruption, fresh-store retry at 20%); VM route failed (TLS-intercepting egress vs Go client); the fleet Azure agent host landed Edge (44 s) + Mini (66 s) 2026-10-29 and Air (22 s + tags) 2026-10-30; registry-side digest verification for all three flagship names; Mini GGUF corrupt-resume catch (sha256) recorded
Gate sanity check	RUN.md; gate49-vendor/VENDOR.md §4	All-decline (bare form) transcript: 23/49, 1 hard failure (pricing) — earlier editions of this register said 0; corrected in v0.4.2 to match the vendor record. Prescribed-decline null: 25/49, 0 hard failures (see served-path register rows below)
Flavor rates	PACK.md; RUN.md	a10g-small $1.00/hr; a10g-large $1.50/hr; wave combined burn $3.50/hr
Publication of this paper (v0.3.0)	HF dataset Thox-ai/thox-whitepapers	Anonymous retrieval verified 2026-10-30: HTTP 200, 508,551 B, SHA-256 fbdf0c63… matches the local v0.3.0 build
Ollama documentation pass	ollama-pages-20261030/TEMPLATE.md; ollama-pages-20261030/INVENTORY.md (program files)	29 public pages inventoried 2026-10-30; 15 completely empty before the pass; TXF-standard READMEs applied 2026-10-30
Launch campaign delivery	model-launch-campaign-20261030/ (program files)	Pack prepared and delivered to Craig Ross, CEO, 2026-10-30; publication is his step
Served-path gate + probe (pre-fix)	Served-path measurement record (RESULTS.md + gate49-A-* / probe-A-* JSONs), 2026-10-30	Azure VM (fleet agent host), Ollama 0.40.13; served digests: Air 5a548903…, Clip 54b3c0e5…, Edge d4f2b564…, Mini 52062e06…; Air 42/49, Edge 16/49 (thinking-surfaced diagnostic 35/49), Mini 46/49, Clip 12/49

Claim class	Source	Identifier
Base baselines (bare + grounded)	Same record (gate49-B-* / gate49-C-* / probe-C-* JSONs)	Qwen2.5-0.5B 4 / 27; Qwen3-0.6B 3 / 25; Qwen3.5-0.8B 0 / 0 (content channel); Phi-4-mini 7 / 27
Template/capability fix + post-fix re-gate	Same record, POST-FIX section (gate49-postfix-* / probe-postfix-* JSONs)	Weights digests unchanged (Clip 1e009136..., Edge 8ef0900d..., Mini a19e9606..., Air fb32e5ab...); Air 42/49, Edge 16/49, Mini 47/49, Clip 40/49
Null-policy measurements	Constant-abstainer report through the vendored scorer (Waves A/B supplement record) + gate49-vendor/VENDOR.md §4	Prescribed form: 25/49, mean 3.531, 0 hard failures, verdict PASS (2026-10-30T10:05:26Z); bare form: 23/49, mean 3.367, 1 hard failure
Grounding-prompt correction	gate49-vendor/VENDOR.md §9; compute-plane Space repos	Pre-correction file 5,032 B (SHA-256 fcd65985...); corrected file 4,645 B, SHA-256 f4bd8a681a531cadea30a6a4e8853804ea3635ff5e965659f9aef3618fdeb54e; commits 635732da / 4c02a4ce (canonical Space), f4587910 (mirror)
License read-back (v0.4.2)	HF API repository file listings, read 2026-10-30	16 distributed repositories checked (§15 table): no LICENSE or NOTICE file in any; card metadata declares the base license in all 16
Post-publication reviews	Research Loop report (whitepaper-review-20261030/research-loop-report.md); Claude hostile review (claude-wavec.md); Imogen packet (email, 2026-10-30T09:55:07Z)	Each finding dispositioned in the Wave C disposition (whitepaper-review-20261030/DISPOSITION-WAVEC.md)

14 Document control

Version	Date	Change
0.1.0	October 28, 2026	Initial draft for founder review. Air and Mini results complete; Edge and Clip sections are pre-registered placeholders pending their in-flight runs. Not submitted or published anywhere.
0.2.0	October 29, 2026	Edge and Clip results filled from their thox_gate49.json reports: §8.3 Edge 46/49, publish bar cleared (the wave's one promotion); §8.4 Clip 43/49, HELD below the bar (known_false repetition, fabricated founding date). Status box, abstract, §8 introduction, Figures 3–4, the Wave C.1 fix program (new item C1-9; retrain set Air + Mini + Clip), Limitations, Conclusion, and the provenance register updated to match. Still a draft for founder review; not submitted or published anywhere.
0.3.0	October 30, 2026	Final edition; program closed. §9 rewritten as the executed fix cycles: Wave C.1 corpus root causes (pricing-form rows; nonexistent-model rows), the termination diagnosis, the Clip C.1 CAO ruling; Wave C.2 termination teaching, the get_device_status tool-name correction, the four-point recipe measurement, Mini's completion-only-loss stack; Air C.2 promoted (47/49) and finished; Clip C.2 (39/49) and Mini C.2 (44/49, probe 5/6) held; the Ollama registry resolution (§9.4); the final program tally (§9.5). §8 rulings carry final-outcome notes; Figures 3–6 regenerated as across-wave history from data/; C.1/C.2 gate and probe reports added to data/; Limitations and Conclusion updated to the closing state. Still a draft for founder review; not submitted or published anywhere.

Version	Date	Change
0.4.0	October 30, 2026	Published final edition. Publication addendum: new §15 (Publication and public documentation) records this paper's public hosting in the Thox-ai/thox-whitepapers dataset, the TXF-standard Ollama documentation pass over the program's model pages, and delivery of the public launch campaign; §13 gains the three corresponding provenance rows. Status changed from draft for founder review to published final edition. The v0.4.0 file is published under its own filename and also replaces the file served at the v0.3.0 address, for link stability; this entry is the supersession record. No result, score, job id, or ruling changed anywhere in the paper.
0.4.1	October 30, 2026	Corrected final edition. Eight corrections from the post-publication Research Loop requirements audit (2026-10-30; report in the internal review record): §5.3 cost paragraph corrected to the attempt record (attempts 1–2 were five failed job instances across four failure classes and failed fast; attempt 3's two jobs ran the full eight-hour timeout, ≈$16 combined, a cost knowingly accepted under the CAO decision of 2026-10-29 00:45 CT); the untraceable "1,159 repositories" survey figure removed from the abstract, §3.1, and §13, the register now citing SURVEY.md's method; §9.2 C.1 corpus accounting reworded (982 − 5 excluded = 977 rows carried forward: 956 verbatim, 21 rewritten, plus 105 added); §11 restores §8.5's confound caveats (upper-bound formulation; no clean corpus-alone or recipe-alone attribution); §2 scopes the QLoRA adapter definition to the plane runs; §9.2 uses "verbatim" for the index-923 match, matching the abstract and RUN.md; §15 scopes the Ollama documentation-pass claim to the flagship and edge pages rewritten and verified in place; §4 qualifies the Early Bird singles as the post-November-2 prices. The v0.4.1 file was published under its own filename and, at the time, also replaced the files served at the v0.4.0 and v0.3.0 addresses for link stability. **That replacement was itself reversed later on October 30, 2026:** the v0.3.0 and v0.4.0 addresses again serve their own editions (508,551 B, SHA-256 fbdf0c63…; 513,921 B, SHA-256 e2cd91f9…), verified by anonymous download, and a `_latest` alias carries the current edition. No result, score, job id, or ruling changed anywhere in the paper.
0.4.2	October 30, 2026	Post-publication revision after review. Integrates the Research Loop audit (already in v0.4.1), the Claude hostile review, and the ChatGPT-lane (Imogen) review packet, each dispositioned in the review record (§13). Changes: authorship and accountability disclosure (front matter); status language corrected to two-of-four promoted, program closed; publish bars restated against the measured null policy with per-category null results (§7.1, §7.3) and the Edge bar's below-null position disclosed; the promotion decision function as actually applied, with all deviations enumerated (§7.5), and §2's hard-failure definition corrected to match it, including the documented-baseline tier; development-set status of the fix-cycle scores disclosed (§7.3); the confounded recipe comparison's stronger claims withdrawn (§8.5, §9.3, §11) — no pure quantization effect is isolated or bounded; new §8.6 reporting served-path GATE49 and probe results for all four shipped tags before and after the template/capability/facts-block fix, the bare and grounded base-model baselines, the Edge thinking-cap finding, and the pending corrected-prompt Modelfile rebake; the shipped Clip weights' training on the twelve later-excluded chipset rows stated (§4); the Air flagship-name window disclosed (§9.4); corpus datasheet corrected (§4); the F2 root cause verified and corrected (§6.2); SmolLM2 training-token figures corrected from the vendor cards (§3.2); the Austin signature/facts-card inconsistency recorded (§7.1); per-artifact license read-back table added (§15); related work added (§12); reproducibility appendix added (Appendix A). No training result, score, job id, or ruling changed.

Version	Date	Change
0.5.0	November 4, 2026	Process v1.0 reissue of v0.4.2: layout moved to the THOX Research Paper Template v1.0; evidence consolidated into a single source (exp/results/summary.json) with every headline figure re-verified against the per-run records (56/56); GATE49 explorer demo added; internal host identifiers genericized to host classes in the paper text and in the staged evidence copies (the already-public HF supplement is untouched). Byline finalized under Tommy Xaypanya's ruling of November 4, 2026: the v0.4.2 authorship of record stands (Sadie Weaving, author; Tommy Xaypanya, accountable human author) and the authorship-and-accountability disclosure added in v0.4.2 is retained verbatim in the front matter. No training result, score, job id, or ruling changed.

15 Publication and public documentation

The program's completion bar treated documentation as part of the deliverable: a model counts as done when it is trained, published, carries a card with its full training record, and is supported by a whitepaper and documentation package. This section records the public-documentation state that closed the program on October 30, 2026. It adds no results; every measurement in this paper stands as reported in Sections 8 and 9.

This paper. Version 0.3.0 of this paper was published on October 30, 2026 in the public Hugging Face dataset Thox-ai/thox-whitepapers, at https://huggingface.co/datasets/Thox-ai/thox-whitepapers/resolve/main/THOX_Wave_C_Flagship_Program_Whitepaper_v0.3.0.pdf, and verified by anonymous retrieval (HTTP 200; SHA-256 fbdf0c63cdc7f0e5476f8b5e3c16d431e05f6da97e87303af6b2786793d4efe0, matching the local build). Versions 0.4.0 and 0.4.1 were each published under their own filenames and, at the time, also replaced the content at the earlier versioned addresses for link stability. That practice was wrong for a paper whose subject is provenance, and it was reversed on October 30, 2026: the v0.3.0 address again serves the v0.3.0 edition (508,551 B, SHA-256 fbdf0c63cdc7f0e5476f8b5e3c16d431e05f6da97e87303af6b2786793d4efe0), the v0.4.0 address the v0.4.0 edition (513,921 B, SHA-256 e2cd91f9a67abd6eac84f240b57057c37a8ba43aa08b3e2bc3c07fcdcf8c8e94), and the v0.4.1 address the v0.4.1 edition (519,308 B, SHA-256 b1f77b3261e2ae6df03b3f8403fbb7df60dee75ebc4e933fc27ce9594f8e8bbf), each verified by anonymous download against the local builds. A separate alias, THOX_Wave_C_Flagship_Program_Whitepaper_latest.pdf, carries the current edition for link stability. This edition, v0.4.2, is published under its own filename and at the alias, and changes no earlier file. (The §14 row for v0.3.0 says "not submitted or published anywhere"; that described the draft's state when the row was written, before the October 30 publication, and §14's v0.4.0 row is the publication record.) The predecessor program paper, *First Device Models, Trained In-House* (Waves A–B, reference 1), was updated to a second edition on the same date, covering the final Wave B results, and is hosted in the same dataset.

Ollama model pages. The program's models are also published through Ollama under the Thox-ai namespace (ollama.com/Thox-ai). An inventory taken on October 30, 2026 counted 29 public model pages in the namespace; 15 of them carried no description and no readme at all, including every flagship push of October 29 — the weights had landed and the pages had not. On October 30, 2026 the flagship and edge model pages were rewritten to a single documentation standard and verified in place — the fleet's TXF-standard template: nine sections comprising a summary, the exact run commands and tags, intended use and device assignment, the model's GATE49 evaluation with its ruling, resources (the Hugging Face repository and this paper), limitations, and license and provenance, with badge images in the brand colour #05A451. The remaining pages in the namespace follow in the same pass, to the same template. The standard does not distinguish by outcome: a held adapter's page carries its held status and its failure record in the same form as a promoted model's page carries its promotion. The template and the inventory are program files (ollama-pages-20261030/TEMPLATE.md and INVENTORY.md).

License compliance on the distributed artifacts (read back 2026-10-30). Earlier editions' Track A language asserted more than the artifacts show. The accurate statement: THOX publishes its fine-tunes under the base models' licenses, with its own contribution offered on those same terms. Verified by read-back of every distributed repository named in this paper: the card metadata declares the base license in all sixteen, and **none of the sixteen carries a LICENSE file or a NOTICE file** — so the MIT notice obligation and the Apache-2.0 §4 license-copy and modification-notice obligations are not discharged inside the artifacts themselves. The gap is recorded here and in the review disposition; adding the files is a finishing-lane task and is not claimed as done in this edition.

Artifact class	Repositories checked	LICENSE file	NOTICE file	Card declares license
Wave C adapters	ThoxAir-0.5B, ThoxMini-Phi4Mini, ThoxEdge-0.8B-WaveC, ThoxClip-Qwen3-0.6B-WaveC	none	none	yes (apache-2.0 / mit)
Fix-cycle adapters	ThoxAir-0.5B-WaveC1, ThoxAir-0.5B-WaveC2, ThoxMini-Phi4Mini-WaveC1, ThoxMini-Phi4Mini-WaveC2, ThoxClip-Qwen3-0.6B-WaveC1, ThoxClip-Qwen3-0.6B-WaveC2	none	none	yes
Merged builds	ThoxAir-0.5B-Merged, ThoxAir-0.5B-WaveC2-Merged, ThoxEdge-0.8B-WaveC-Merged, ThoxMini-Phi4Mini-Merged	none	none	yes
Prior published	ThoxClip-Qwen3-0.6B, ThoxEdge-0.8B	none	none	yes (apache-2.0)

Reproducibility supplement. A supplement to this edition is published in the same dataset under the `wavec-supplement/` prefix: the vendored GATE49 bundle (scorer, facts package, facts card v1.1.0, grounding prompt as corrected, probe), the per-run gate and probe reports for every program run, the served-path and baseline reports of

§8.6 (pre- and post-fix), the null-policy reports, the registry entries the runs were launched from, and a manifest with SHA-256 hashes. Appendix A states what the supplement does and does not make reproducible.

Launch campaign. A public model-launch campaign announcing the models' availability was prepared — posts for X, LinkedIn, Instagram and Facebook, a Hugging Face announcement, a Kickstarter backer update, a short announcement video script, and graphics — and delivered to Craig Ross, Chief Executive Officer, for publication on October 30, 2026. Under the company's social process, publication is the CEO's step; this paper records the delivery, not the posting.

A Appendix A. Reproducibility (added in v0.4.2)

The gate package. The exact instrument that produced every GATE49 number in this paper is the vendored bundle in the supplement (`wavec-supplement/gate49/`): `factuality_gate49.py` (the fleet scorer, differing from its factory original by one REPO_ROOT line, recorded in the bundle's VENDOR.md), the `facts` package, the facts card v1.1.0 (`facts/data/thox_facts.json`, SHA-256 `635d81ba119089328aca60bab11b5285347f0a73fa81cda686ff5328a8a163c9`), `grounding_system.txt` (corrected version, 4,645 B, SHA-256 `f4bd8a681a531cadea30a6a4e8853804ea3635ff5e965659f9aef3618fdeb54e`; the probe runs and the shipped Modelfiles used the pre-correction 5,032 B version, SHA-256 `fcd65985…`, §8.6), `run_gate49_job.py`, and `probe_flagship.py`. The bundle is committed in both compute-plane repositories (canonical commit `8b46f482facf4a22902f6cddfbba125b1c1a12d8`; mirror `064f5729818f0a9680de86108c528d62aa69f34b`).

Full hashes. Corpus files: Wave B `6665a8cbd79197c61a9db8119b743faedc3e469407364bb553ec9478e90a5784`; Wave C.1 `da707b20380e64f3346889f89e0137d06719d2d0f126bd6ece7a4edf017e24c2`; Wave C.2 `771434812b6a14fceba7f929fcbe1040d0e6be7a32c9495c1e8d0784512919a3`; predecessor `ac118a250bc989c9478ec7e4ebdcc706e12e489979b02e6bf1af8b831f8b2514`. Promoted Air C.2 merged weights (988,097,824 B): `2497a347a4ac0a9b92e6e6be50ed6fb5b2c7d1bb2c174b9c1f1aa04f3e1433fb`; its Q4_K_M GGUF and Ollama layer: `fb32e5ab802df466809836182f071bb0ab0871abb9273131456c89d998bccd6a`. Edge Ollama layer: `8ef0900d1f76da67442dca354a778517ddc008b9f73926df874f7f3dff5d5625`. Mini Ollama layer: `a19e96061d84ca2014d4de98d0c0397e6df4db8f73c2163afa6bcb90d15afedf`. Served-run model digests are in the measurement record (§13).

Per-run recipe table. "Loss" is the prompt-masking behaviour of the run's stack generation (§6.1). "Final train loss" figures in §8 and §9 are the HF Trainer's run-average `train_loss` as recorded in the job records, not last-step losses; cross-tokenizer comparisons of those numbers are not meaningful.

Run	Route	Precision	Loss	Steps
Air Wave C	plane (profile `latest`)	QLoRA NF4	completion-only	186
Mini Wave C	plane (profile `pinned`)	QLoRA NF4	full-prompt	186
Edge Wave C	plane (fleet pins, pytorch image)	QLoRA NF4	completion-only	186
Clip Wave C	plane (fleet pins, pytorch image)	QLoRA NF4	completion-only	186

Run	Route	Precision	Loss	Steps
Air C.1	plane	QLoRA NF4	completion-only	204
Mini C.1	pinned stack	QLoRA NF4	full-prompt	204
Clip C.1	plane	QLoRA NF4	completion-only	204
Air C.2	custom job, `thox-edge-sft` @ `c250f9b`	fp-LoRA	completion-only	468
Mini C.2	custom stack (transformers 4.53.3 / trl 0.20.0)	QLoRA NF4	completion-only	234
Clip C.2	plane	QLoRA NF4	completion-only	234

Decoding and seeds. Gate and probe generation is greedy, max 320 new tokens, weights-only house prompt for the gate and the grounding prompt for the probe. No training seed is recorded in any registry entry or job record in this program; runs are identified by job id. Exact re-execution would require re-derivation from the registry entries and profiles in the supplement; bit-exact reproduction is not claimed.

What the supplement does not contain. The training corpora themselves remain in the private dataset repository; their hashes, composition, changelogs, and row-class records are published (here and in the supplement manifest), and the gate package is complete — so every *score* in this paper can be re-derived against any candidate artifact, while re-running the *training* requires the private corpora. Wave A/B per-question gate outputs were not retained on the run hosts (recorded in that program's supplement) and cannot be reconstructed; all Wave C per-run reports exist and are included.

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